

Galaxy mergers in the epoch of reionization – II. Merger-triggered star formation and AGN activities at $z = 4.5$ – 8.5

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ABSTRACT

Galaxy mergers are key drivers of galaxy formation and evolution, including triggering active galactic nucleus (AGN) and star formation to a still unknown degree. We thus investigate the impact of galaxy mergers on star formation and AGN activity using 3330 galaxies at $z = 4.5$ – 8.5 from eight *James Webb Space Telescope* fields (CEERS, JADES GOODS-S, NEP-TDF, NGDEEP, GLASS, El-Gordo, SMACS-0723, and MACS-0416), covering an unmasked area of 189 arcmin². We focus on star formation rate (SFR) enhancement, AGN fraction, and AGN excess in close pairs defined by $\Delta z < 0.3$ and projected separations $r_p < 100$ kpc, relative to non-merger samples. Close pairs with mass ratios greater than 1:4 are used for the SFR-enhancement analysis, whereas no mass-ratio constraint is applied for the AGN fraction and AGN excess measurements. We find SFR enhancement occurs only at $r_p < 20$ kpc, with values 0.25 ± 0.10 dex and 0.26 ± 0.11 dex above non-merger medians for $z = [4.5, 6.5]$ and $z = [6.5, 8.5]$, respectively. No other statistically significant enhancements in galaxy sSFR or stellar mass are observed at any projected separation or redshift. We also compare observational results with predictions from the SC-SAM simulation, finding no evidence of star formation enhancement in simulations at any separation. Lastly, we examine the fraction and excess of AGNs identified through photometric spectral energy distributions (SED) fitting (Type-I) and BPT diagnostics (Type-II). We find $48^{+33}_{-7}\%$ of Type-I and $44^{+26}_{-20}\%$ of Type-II AGNs have a close companion within $r_p < 50$ kpc and $\Delta z < 0.3$. Furthermore, $73^{+27}_{-32}\%$ of AGNs have companions within 100 kpc. Relative to isolated galaxies, we measure an AGN excess factor of $1.26^{+0.15}_{-0.04}$ for Type-I and $1.34^{+0.23}_{-0.11}$ for Type-II AGNs in close pairs, suggesting notable AGN enhancement in galaxy pairs at these higher redshifts.

Key words: galaxies: evolution – galaxies: formation – galaxies: high-redshift – galaxies: interactions – galaxies: Seyfert – galaxies: star formation.

1 INTRODUCTION

Galaxy mergers are a fundamental mechanism driving the evolution of galaxies. Theory and simulations show that tidal interactions from mergers induce bar instabilities in the centres of galaxies, leading to strong gravitational torques that funnel gas towards the central regions, triggering both starbursts and active galactic nucleus (AGN) activity (e.g. J. E. Barnes & L. Hernquist 1996; J. C. Mihos & L. Hernquist 1996; T. J. Cox et al. 2006; P. Di Matteo

et al. 2007; P. F. Hopkins et al. 2008; M. Montuori et al. 2010; D. S. Rupke, L. J. Kewley & J. E. Barnes 2010; J. M. Scudder et al. 2012; P. Torrey et al. 2012a; J. Moreno et al. 2015, 2019).

Observational studies have provided crucial insights into how star formation evolves during mergers, but a comprehensive understanding of the temporal and environmental factors influencing these changes is still developing. In particular, there is a need to trace variations in the star formation rate (SFR) across different stages of mergers and at various cosmic epochs to quantify the impact of these interactions on galactic evolution. Studies at low redshifts ($0 < z < 1$) have shown significant SFR enhancements in galaxy pairs. Using data from the Sloan Digital Sky Survey (SDSS),

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multiple studies (e.g. S. L. Ellison et al. 2008, 2010; D. R. Patton et al. 2011; J. M. Scudder et al. 2012; D. R. Patton et al. 2013, 2016) have reported that galaxies in close pairs (with projected separations of less than 40–80 kpc) exhibit, on average, a 60 per cent higher SFR compared to controlled isolated galaxies. Similarly, L. E. Garduño et al. (2021), using data from the Galaxy and Mass Assembly (GAMA) survey (S. P. Driver et al. 2011), found SFR enhancements ranging from 0.025 to 0.15 dex. At higher redshifts, C. N. Lackner et al. (2014) identify merging galaxies in the COSMOS field between $0.25 < z < 1.00$ at projected separations of $r_p = 2.2\text{--}8$ kpc, reporting an SFR enhancement of 2.1 ± 0.6 in their merging sample compared to non-merging galaxies. Similarly, E. A. Shah et al. (2022) found an SFR enhancement factor of $\sim 1.23^{+0.08}_{-0.09}$ in the redshift range $0.5 < z < 3$. Besides the aforementioned literature, there are several works who have verified the SFR enhancement in close-pairs (e.g. L. J. M. Davies et al. 2015; J. Moreno et al. 2021; P. Calderón-Castillo & R. Smith 2024).

However, there are studies which have found the opposite of this. In contrast to induced star formation, A. Silva et al. (2018) find no significant difference in star formation properties between merger and non-merger galaxies at $0.5 < z < 2.5$ in the projection separation range of $3 < r_p < 15$. W. J. Pearson et al. (2019), using convolutional neural networks to detect binary mergers in SDSS (DR7), KiDS (J. T. A. Jong et al. 2012), and CANDELS (N. A. Grogin et al. 2011; A. M. Koekemoer et al. 2011) imaging, reported only a 20 per cent increase in SFR due to merger activity. Additionally, S. L. Ellison et al. (2022) found that post-merger galaxies are 30–60 times more likely to rapidly shut down their star formation after a starburst than a control sample of non-merging galaxies, a phenomenon that might explain the quenching of galaxy GS-9209 reported by R. G. Pascalau et al. (2025). Interestingly, no such excess shutdown was found among close galaxy pairs, suggesting that while mergers can trigger star formation, they can also lead to its rapid cessation, particularly after the merging process is complete. Y. A. Li, L. C. Ho & J. Shangguan (2023), analysing SDSS data within the redshift range $0.01 < z < 0.11$, find that in low-redshift galaxies, mergers – including those between gas-rich and nearly equal-mass galaxies – have a negligible effect on their SFR, specific SFR, and star formation efficiency. Starbursts remain uncommon, and the star formation efficiency of gas-rich minor mergers may even be reduced.

There are two primary observational approaches to investigate the impact of galaxy interactions on galaxy star-forming properties. One approach uses morphological parameters, deep learning, and convolutional neural network methods on galaxy morphology to identify merging galaxies and compare their star-forming properties with controlled non-merging galaxies (e.g. C. N. Lackner et al. 2014; A. Silva et al. 2018; W. J. Pearson et al. 2019; N. Dalmasso et al. 2024; L. Ferreira et al. 2025). The other approach is the close-pair method, which investigates the extent of SFR enhancement at various projected separations to the closest neighbour (e.g. S. L. Ellison et al. 2010; J. M. Scudder et al. 2012; E. A. Shah et al. 2022). The morphological method is influenced by various intrinsic and extrinsic galaxy properties, such as orbit type, gas fraction, viewing angle, wavelength, redshift, and depth (J. M. Lotz et al. 2011; K. A. Blumenthal et al. 2020), which introduces challenges in identifying merging galaxies at the very high-redshifts ($z = [4.5, 8.5]$) used in this study. Therefore, we conduct an early *James Webb Space Telescope* (JWST) analysis using the

close-pair method to investigate the enhancement of galaxy star-forming properties for the most distant galaxies.

Besides triggering star formation, galaxy mergers can also drive rapid gas accretion on to supermassive black holes (SMBHs), thereby initiating AGN activity (e.g. E. Koulouridis et al. 2006a, b; S. L. Ellison et al. 2011; E. Koulouridis et al. 2013; E. Koulouridis 2014; S. Satyapal et al. 2014; R. S. Barrows et al. 2017; C. Ricci et al. 2017; F. Gao et al. 2020; B. L. C. Araujo et al. 2023; S. Byrne-Mamahit et al. 2023; J. C. S. Pierce et al. 2023). Inflowing material can obscure the central regions, making the most heavily obscured AGNs elusive to optical and soft X-ray surveys. Infrared observations offer a more reliable detection method due to their lower susceptibility to absorption (C.-Y. Chiang et al. 2019). With the high resolution of the JWST NIRSpec instrument, AGNs can be identified either through the broad-line components of key emission lines ($H\alpha$, $H\beta$) (Y. Harikane et al. 2023; R. Maiolino et al. 2024, 2025) or using diagnostic diagrams (L. J. Kewley et al. 2001; G. Kauffmann et al. 2003; L. J. Kewley et al. 2006; M. Perna et al. 2023; J. Scholtz et al. 2023; G. Mazzolari et al. 2024). Photometrically, AGN detection can also be inferred from SED templates (K. Nakajima & R. Maiolino 2022; I. Juodžbalis et al. 2023).

Understanding the fraction of galaxies that harbour AGNs is crucial for grasping the black hole accretion history of the Universe. J. M. Comerford et al. (2024) analysed MaNGA data and identified an AGN excess of about 1.8 in major mergers and 1.5 in minor mergers relative to non-merger isolated samples ($z < 0.15$). F. Gao et al. (2020) investigate AGN fractions using low-redshift SDSS samples at $0.005 < z < 0.1$ (K. N. Abazajian et al. 2009) and slightly higher redshift samples from GAMA ($0 < z < 0.6$) (S. P. Driver et al. 2011; J. Liske et al. 2015). Their study revealed a higher AGN fraction in merger samples compared to non-merger samples. Specifically, they report AGN fractions of 1.63 ± 0.05 per cent versus 1.45 ± 0.02 per cent in optical and 0.34 ± 0.02 per cent versus 0.20 ± 0.01 per cent in mid-infrared (MIR) for SDSS AGNs and controls, respectively. Similarly, in GAMA, the optical AGN fractions are found to be 3.63 ± 0.11 per cent for mergers versus 2.53 ± 0.03 per cent for the controls; within the MIR the fractions are 0.73 ± 0.05 per cent versus 0.53 ± 0.01 per cent. Furthermore, at intermediate redshifts ($0.5 < z < 2.5$), using X-ray and UV-to-far-infrared data, T. Wang et al. (2017) found that red galaxies exhibit the lowest AGN fraction (~ 5 per cent) at $z \sim 1$, but show the most rapid evolution with redshift, increasing by a factor of ~ 5 to reach ~ 24 per cent at $z \sim 2$. Green valley galaxies display the highest AGN fractions across all redshifts, peaking at $z \sim 2$, where more than half host an AGN at $M_* > 10^{10.6} M_\odot$. In addition, numerous studies have shown that the AGN fraction increases as the separation between galaxy pairs decreases (S. L. Ellison et al. 2011; J. Silverman et al. 2011; M. Koss et al. 2012; S. L. Ellison et al. 2013; E. T. Khabiboulline et al. 2014; S. Satyapal et al. 2014; R. W. Bickley et al. 2022; R. W. Bickley et al. 2024).

Most of the literature on merger triggered star formation and AGN activity focuses on redshifts $z < 4.5$. The process at $z > 4.5$ epoch remains largely, if not completely, unknown. The high-redshift Universe is increasingly unveiled by JWST, highlighted by numerous recent studies on early galaxy discoveries (e.g. H. Atek et al. 2022; M. Castellano et al. 2022; E. Curtis-Lake et al. 2022; C. T. Donnan, R. Tojeiro & K. Kraljic 2022; Y. Harikane et al. 2022; R. P. Naidu et al. 2022; N. J. Adams et al. 2023; R. J. Bouwens et al. 2023; C. M. Casey et al. 2023; C. Donnan et al. 2023; S. L. Finkelstein et al. 2023; M. Franco et al. 2023; K. N. Hainline

et al. 2023; R. Maiolino et al. 2023; D. J. McLeod et al. 2023; P. Arrabal Haro et al. 2023a; D. Austin et al. 2023a; N. J. Adams et al. 2024; P. Chakraborty et al. 2024; R. Maiolino et al. 2024; C. J. Conselice et al. 2024b). These studies reveal a significant population of distant galaxies at $z > 4.5$, previously underrepresented in *Hubble Space Telescope* (*HST*) data, providing a new window to explore merger evolution at high redshifts. This paper is the second in the *Galaxy Mergers in the Epoch of Reionization* series. The first paper in this series (Q. Duan et al. 2025) presented, for the first time, galaxy pair fractions, merger rates, and stellar mass accretion rates from mergers at $z = 4.5$ – 11.5 . It revealed a rapid merger rate of approximately 6 mergers per Gyr at these redshifts, with mergers contributing 71 ± 25 per cent of the stellar mass growth in galaxies, comparable to the increase from star formation fuelled by gas. In this paper, we extend this work to explore the impact of mergers on galaxy star formation and AGN activity at $z = 4.5$ – 8.5 , using the same data as the first study from eight *JWST* observational fields: CEERS, JADES GOODS-S, NEP-TDF, NGDEEP, GLASS, El-Gordo, SMACS-0723, and MACS-0416.

The structure of this paper is as follows. In Section 2, we describe the data we use and outline our methods for redshift measurement, SFR and stellar mass inference. Our approach for classifying close-pair galaxies is detailed in Section 3. The main findings are presented in Section 4. Finally, we discuss our results in Section 5 and summarize our conclusions in Section 6. Throughout this work, we adopt the Planck 2018 cosmology (Planck Collaboration VI 2020), with $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_M = 0.315 \pm 0.007$, and $\Omega_\Lambda = 0.685 \pm 0.007$, to ensure consistency with other observational studies. All magnitudes reported are in the AB magnitude system (J. B. Oke 1974; J. B. Oke & J. E. Gunn 1983).

2 OBSERVATIONS, DATA REDUCTION, AND DATA PRODUCTS

The launch of the *JWST* in December 2021 (J. Rigby et al. 2023) has ushered in a new era of high-redshift Universe exploration. In this work, we consistently reduce and analyse publicly available deep *JWST* NIRCam data from the PEARLS, CEERS, GLASS, JADES GOODS-S, NGDEEP, and SMACS0723 surveys. Specifically, for the PEARLS programme, we include data from the NEP-TDF, MACS-0416, and El-Gordo fields, resulting in a combined total of 189 arcmin² of unmasked sky when integrated with the other five surveys. From this extensive dataset, we have identified 2404 robust galaxy candidates at $z_{\text{phot}} = [4.5, 6.5]$, with 1276 candidates at $z_{\text{phot}} > 6.5$ (926 at $z_{\text{phot}} = [6.5, 8.5]$), forming our EPOCHS V1 sample (C. J. Conselice et al. 2024a). For the $z_{\text{phot}} = 4.5$ – 6.5 range, we utilize data only from the CEERS, JADES GOODS-S, and NEP-TDF surveys due to the availability of *HST* ACS/WFC *F606W* and *F814W* band data.

For the SMACS-0723, MACS-0416, and El-Gordo cluster observations, one NIRCam module is centred on the lensing cluster in each pointing, while the other module is positioned approximately 3 arcmin away, providing an effectively ‘blank-field’ view of the distant Universe. In this paper, we exclude sources found in the cluster module to avoid potential uncertainties in corrections for mergers and contamination from cluster members, as well as complications arising from lensing effects (e.g. A. Griffiths et al. 2018; R. Bhatawdekar & C. J. Conselice 2021). For these clusters, only the NIRCam module that is not centred on the cluster is used.

We utilize our own EPOCHS reduction pipeline, with details provided in N. J. Adams et al. (2024), D. Austin et al. (2024), T. Harvey et al. (2024), and C. J. Conselice et al. (2024a). While we do not repeat the full details here, the methods are thoroughly discussed in these papers. It is essential to note that all fields are reprocessed using the same methodology, ensuring uniformity in galaxy detection and measurement. We have reprocessed all lower-level *JWST* data products using a modified version of the official *JWST* pipeline. For further details on the data reduction process and catalogues, please refer to N. J. Adams et al. (2024), D. Austin et al. (2024), T. Harvey et al. (2024), and C. J. Conselice et al. (2024a).

In the following subsections, we describe the various fields studied and the relevant properties for this paper. The depth of our data, collected in various fields and *JWST* filters, as well as in *HST* ACS, is outlined in table 1 of N. J. Adams et al. (2024) within C. J. Conselice et al. (2024b).

2.1 CEERS

The Cosmic Evolution Early Release Science Survey (CEERS; PI: S. Finkelstein, PID: 1345, M. B. Bagley et al. 2023b) is one of the 13 Early Release Science (ERS) programmes conducted during the first year of *JWST*’s operation. CEERS features a $\sim 100 \text{ arcmin}^2$ NIRCam imaging mosaic covering wavelengths from 1 to 5 microns, across 10 NIRCam pointings. Concurrently, CEERS includes imaging with the mid-infrared instrument (MIRI) over eight pointings, covering 5 to 21 microns, and multislit spectroscopy using the near-infrared spectrograph (NIRSpec) across six pointings. NIRCam and *HST* imaging data is used in this work. This includes imaging across nine distinct filters: *F606W*, *F814W*, *F115W*, *F150W*, *F200W*, *F277W*, *F356W*, *F410M*, and *F444W*, with a 5σ depth of 28.6 AB magnitudes using 0.16 arcsec radius circular apertures.

In addition to photometric data, we also use CEERS Spectroscopic data. The CEERS NIRSpec spectroscopic data (S. Fujimoto et al. 2023; P. Arrabal Haro et al. 2023a, b) were procured as part of the ERS programme (PI: Steven L. Finkelstein, ID:1345). This data set was designed to optimize the overlap with observations from both NIRCam and *HST*, using three medium resolution gratings $R \approx 1000$ and the PRISM $R \approx 100$. During this period, both NIRSpec pointings, namely NIRSpec11 and NIRSpec12, adhered to the standard CEERS MSA observational guidelines. Specifically, they encompassed three integrations with 14 groups in the NRSIRS2 (Near-Infrared Spectrograph Improved Reference Sampling and Subtraction version 2) readout mode per visit, leading to a total exposure time of 3107 s. Within these observations a trio of shutters was used to form slitlets, facilitating a three-point nodding sequence to enhance background subtraction. The PRISM disperser, ranging in wavelength from 0.6 to 5.3 μm , is characterized by its capacity to provide varied spectral details. In this paper, we use the NIRSpec data reduced by the Cosmic Dawn Center, which is published on the DAWN *JWST* Archive (DJA).¹

¹<https://dawn-cph.github.io/dja/blog/2023/07/18/nirspec-data-products/>.

2.2 JADES Deep GOODS-S

The *JWST* Advanced Deep Extragalactic Survey (JADES; A. J. Bunker et al. 2023; D. J. Eisenstein et al. 2023; K. Hainline et al. 2023; M. Rieke & others 2023; F. D'Eugenio et al. 2024) covers both the GOODS-S and GOODS-N fields. In this paper, we focus on the region covered by the JADES DR1 release that is in the GOODS-S field (PI: Eisenstein, N. Lützgendorf, ID:1180, 1210). The observations utilize nine filter bands: *F090W*, *F115W*, *F150W*, *F200W*, *F277W*, *F335M*, *F356W*, *F410M*, and *F444W*, encompassing a spatial extent of 24.4–25.8 arcmin². A minimum of six dither points was used for each observation, with exposure times spanning 14–60 ks. The depth of the JADES data within the *JWST* bands ranges from 29.58 to 30.21, with the deepest band *F277W*. We also use the *HST* reductions from v2.5 of the Hubble Legacy Fields Project (G. Illingworth et al. 2016; K. E. Whitaker et al. 2019), including ACS/WFC3 *F606W* and *F814W* filter band data.

We also incorporate public JADES NIRSpec data. We use the third JADES released NIRSpec (P. Ferruit et al. 2022; A. J. Bunker et al. 2023; F. D'Eugenio et al. 2024) data (PI: Eisenstein, N. Lützgendorf, ID:1180, 1210), with a focus on the publicly released data in GOODS-S field. The spectra are obtained through the application of both disperser/filter and PRISM/clear configurations. Four different disperser/filter combinations are used to acquire the spectroscopy: G140M/F070LP, G235M/F170LP, G395M/F290LP, and G395H/F290LP, with a wavelength coverage of 0.70–1.27, 1.66–3.07, 2.87–5.10, and 2.87–5.14 μ m, respectively. The three medium-resolution filters have a nominal resolving power of $R \approx 1000$, while the high-resolution data can reach $R \approx 2700$. In this paper, we primarily utilize the PRISM data, which covers a wavelength range of 0.6–5.3 μ m, and exhibits a spectral resolution of $R \approx 30$ –330 (Z. Ji & M. Giavalisco 2022).

2.3 PEARLS and NEP-TDF

The North Ecliptic Pole Time-Domain Fields (NEP-TDF) is part of the *JWST* PEARLS observational programme. The Prime Extragalactic Areas for Reionization and Lensing Science project (PEARLS; PIDs 1176, 2738, PI: Rogier Windhorst; J. M. Diego et al. 2023; B. L. Frye et al. 2023; R. A. Windhorst et al. 2023) is a *JWST* Guaranteed Time Observation (GTO) programme. As a Cycle 1 GTO project, the PEARLS team was allocated a science time of 110 hours. The project's primary objective is to capture medium-deep NIRCам imaging of blank and cluster fields with an approximate depth of 28–29 AB magnitudes.

In this study, we use the first two lensing clusters observed by PEARLS, MACS-0416 and El Gordo, along with NEP-TDF. The data acquired from PEARLS includes imaging in seven wide filter bands: *F090W*, *F115W*, *F150W*, *F200W*, *F277W*, *F356W*, and *F444W*, along with one medium band: *F410M*. In addition, we also include bluer *F606W* imaging from the *HST* Advanced Camera for Surveys (ACS) Wide Field Channel (WFC) in the NEP-TDF from the GO-15278 (PI: R. Jansen) and GO-16252/16793 (PIs: R. Jansen & N. Grogin, see R. O'Brien et al. 2024) *HST* programmes.

For the lensing fields MACS 0416, and El Gordo, the observations include pointings with one NIRCам module centred on the lensing cluster, while the second module is offset by around 3 arcminut in a 'blank' region. Although we process both modules in these fields, we do not include sources found in the module containing the lensing cluster in this study.

2.4 NGDEEP

The Next Generation Deep Extragalactic Exploratory Public Survey (NGDEEP; ID 2079, PIs: S. Finkelstein, Papovich, and Pirzkal) (G. C. K. Leung et al. 2023; D. Austin et al. 2023a; M. B. Bagley et al. 2023a) is the deepest public survey with *JWST* NIRCам conducted in the first year of observations. Initially scheduled for late January to early February 2023, NGDEEP's primary focus involved NIRISS Wide Field Slitless Spectroscopy of galaxies within the Hubble UltraDeep Field. However, due to an unforeseen suspension of NIRISS operations during the survey's observation window, only half of the planned observations were executed, with the remainder taken in early 2024. In this work we use exposures from the second epoch of observations. The parallel NIRCам observations from NGDEEP encompass six wide-band photometric filters (*F115W*, *F150W*, *F200W*, *F277W*, *F356W*, *F444W*). In addition, we also use *HST* data from the same Hubble Legacy Fields project as JADES. Some of our early work in this field was presented in D. Austin et al. (2023b).

2.5 GLASS

The Grism Lens Amplified Survey from Space (GLASS) observational programme (GLASS; PID 1324, PI: T. Treu; T. Treu et al. 2022) is centred on the Abell 2744 Hubble Frontier Field lensing cluster. GLASS primarily employs NIRISS and NIRSpec spectroscopy to study galaxies lensed by the Abell 2744 cluster. In parallel, NIRCам is used to observe two fields offset from the cluster centre. Due to the minimal lensing magnification in these offset areas (P. Bergamini et al. 2023), they effectively serve as blank fields. The observations include seven wide filter bands: *F090W*, *F115W*, *F150W*, *F200W*, *F277W*, *F356W*, and *F444W*.

2.6 SMACS J0723-7327 ERO

The SMACS-0723 lensing cluster is part of the very first ERO programme (PID 2736, PI: Klaus Pontoppidan, K. M. Pontoppidan et al. (2022)). The programme includes NIRCам, MIRI, NIRISS, and NIRSpec observations, and we use the NIRCам observations which contains six bands: *F090W*, *F150W*, *F200W*, *F277W*, *F356W*, and *F444W*. The exposure time of this observation is set to roughly reach the Frontier depth in *F814W*, 1.5 times the *F160W* depth, and almost 10 times the Spitzer (M. W. Werner et al. 2004) IRAC 1 + 2 depth. Because of the inaccuracies of the photometric redshifts for galaxies within this field around $z \sim 10$ due to the lack of the *F115W* filter band, for the most part the pairs in this field are at the lower or higher redshifts of our range.

2.7 Photometric redshift measures

The photometric redshift measurements and selection criteria for our EPOCHS samples have been discussed in detail in the EPOCHS series papers. In this paper, we summarize the method. For an in-depth explanation, we refer readers to N. J. Adams et al. (2024), D. Austin et al. (2024), T. Harvey et al. (2024), and C. J. Conselice et al. (2024a).

We use EAZY-PY (hereafter EAZY (G. B. Brammer, P. G. Dokkum & P. Coppi 2008)) as our primary photometric SED-fitting code. We employ the 'tweak_fsps_QSF_12_v3' templates alongside Sets 1 and 4 of the SED templates developed by R. L. Larson et al. (2023). These additional templates are optimized for high-redshift galaxies, which exhibit bluer rest-frame UV colours

(M. W. Topping et al. 2022; F. Cullen et al. 2023; T. Nanayakkara et al. 2023; D. Austin et al. 2024; F. Cullen et al. 2024) and stronger emission lines (e.g. S. Withers et al. 2023), characteristics frequently observed in such young and distant galaxies.

The ‘tweak_fsps_QSF_12_v3’ templates are generated using the flexible stellar population synthesis (FSPS) package (C. Conroy & J. E. Gunn 2010). Set 1 includes three pure stellar templates from R. L. Larson et al. (2023), which are based on the BPASS v2.2.1 stellar population synthesis (SPS) model (J. J. Eldridge et al. 2017; E. R. Stanway & J. J. Eldridge 2018; C. M. Byrne et al. 2022) for ages of 10^6 , $10^{6.5}$, and 10^7 years at a fixed metallicity $Z_* = 0.05 Z_\odot$, assuming a standard Chabrier (G. Chabrier 2003) IMF. Set 4 incorporates nebular continuum and line emission calculated using CLOUDY v17 (G. J. Ferland et al. 2017), with an ionization parameter $\log U = -2$, gas metallicity $Z_{\text{gas}} = Z_*$, Lyman continuum escape fraction $f_{\text{esc, LyC}} = 0$, hydrogen density $n_{\text{H}} = 300 \text{ cm}^{-3}$, and a spherical geometry, excluding Ly α emission.

We execute the EAZY-PY fitting three times: first with a uniform redshift prior of $0 \leq z_{\text{phot}} \leq 25$, and subsequently with a constrained upper redshift limit of $z_{\text{phot}} \leq 4$ and $z_{\text{phot}} \leq 6$. This triple approach allows us to compare the goodness of fit between high and low-redshift solutions, as indicated by the χ^2 values, ensuring robust identification of high-redshift galaxies. The typical photometric redshift 1σ uncertainty from EAZY-PY is approximately 0.12.

To account for uncertainties in flux calibrations and aperture corrections, as well as potential mismatches between synthetic templates and observed galaxies, we impose a minimum flux uncertainty of 10 per cent (M. J. Rieke et al. 2023). Our selection criteria rely predominantly on specific cuts in computed quantities to minimize biases and incompleteness. Although we perform a visual review of cut-outs and SED-fitting solutions, this step results in less than 5 per cent rejection of our original sample.

2.8 Star formation rate and stellar mass measures

To infer galaxy stellar masses from photometry, we use the Bagpipes (A. Carnall et al. 2018) Bayesian SED-fitting code, adopting redshift probability distribution functions from for each galaxy. We use logarithmic priors for dust, metallicity, and age. The reason for selecting logarithmic priors is because we expect high redshift galaxies to be young, with a lower metallicity and to contain little dust. We set prior limits for metallicity in the range of $[5e-03, 5] Z_\odot$, and we use a dust prior in the range of $[0.0001, 10.0]$ in A_V , ionization parameter $\log_{10}(U)$ in the range of $[-4, -1]$. Bagpipes does not allow SFH to extend beyond τ_U , the age of the Universe at that redshift. In addition, P. Kroupa (2001) IMF, G. Bruzual & S. Charlot (2003) SPS model, and the D. Calzetti et al. (2000) dust attenuation model are implemented (R. A. A. Bowler et al. 2022).

We fit the galaxy photometry with six parametric SFH models – lognormal, delayed, constant, exponential, double delayed, and delayed burst – along with a non-parametric Continuity model (J. Leja et al. 2019). These models have been used in various works (e.g. A. C. Carnall et al. 2023; R. Endsley et al. 2023; L. Whitler et al. 2023; S. Tacchella et al. 2023a). Different SFHs can have a large impact on inferred stellar mass and SFR (L. J. Furtak et al. 2021; S. Tacchella et al. 2022; T. Harvey et al. 2024; B. Wang et al. 2024). We have chosen to present results inferred from our fiducial model, the lognormal SFH with 10Myr averaged time-scale, because we expect high-redshift galaxies to exhibit rising

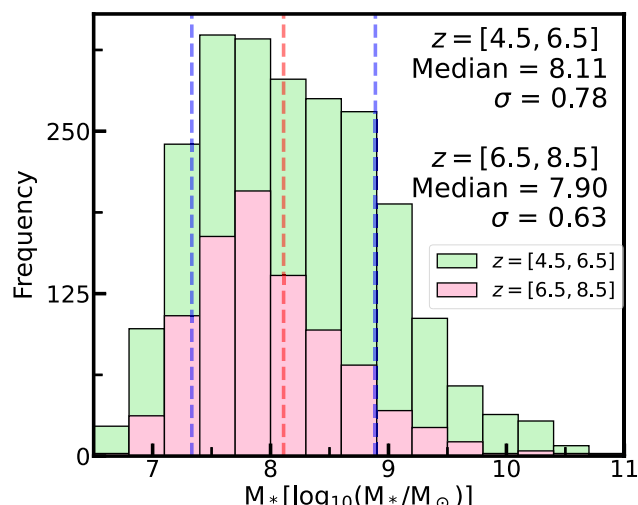


Figure 1. Histogram displaying the distribution of the best-fitting stellar masses for all 2404 galaxies at $z = [4.5, 6.5]$ and for 926 galaxies at $z = [6.5, 8.5]$. These samples are from our eight *JWST* fields: CEERS, JADES GOODS-S, NEP-TDF, NGDEEP, GLASS, El-Gordo, SMACS-0723, and MACS-0416. The median stellar masses are found to be $\sim 8.1 [\log_{10}(M_*/M_\odot)]$ and $7.9 [\log_{10}(M_*/M_\odot)]$ for each redshift bin respectively. As can be seen, it is clear that there is a drop in completeness at $\log_{10}(M_*/M_\odot) < 8$ which is also seen in image simulations of these fields (T. Harvey et al. 2024).

or bursting star formation (Q. Duan et al. 2024). Please refer to T. Harvey et al. (2024) for a thorough discussion of the topic of priors in SED fitting.

We show the stellar mass distribution of the 3300 galaxies used in this work in Fig. 1. The green histogram represents the distribution of 2404 galaxies at $z = [4.5, 6.5]$, while the pink histogram shows the distribution of 926 galaxies at $z = [6.5, 8.5]$. In both redshift bins, the median stellar mass is approximately $\log_{10}(M_*/M_\odot) \sim 8$. Due to the depth limitations of our survey, many low-mass, faint galaxies are missed. Consequently, the number of galaxies below $\log_{10}(M_*/M_\odot) \sim 8$ is significantly reduced and incomplete. At the high-mass end, the number of detected galaxies beyond $\log_{10}(M_*/M_\odot) = 10$ is too small for meaningful statistical analysis. Therefore, to ensure a robust and consistent sample, we limit our analysis to galaxies with stellar masses in the range $8 < \log_{10}(M_*/M_\odot) < 10$, resulting in a total of 1569 galaxies.

3 CLOSE-PAIR CLASSIFICATIONS

We have two primary objectives in this work: the first is to identify close-pair merging systems and analyse how SFR, specific SFR (sSFR), and stellar mass change with proximity to the nearest companion. The second is to investigate AGN Fraction and AGN Excess in these systems. Having well-defined criteria for identifying close-pair samples is crucial, as it directly impacts the reliability of our results (Q. Duan et al. 2025). In this section, we outline the criteria used to classify galaxies into close-pair *major merger* and non-merger systems.

To be classified as part of the close-pair *major merger* sample, a galaxy must simultaneously satisfy all three criteria outlined below. Galaxies that fail to meet any one of these criteria are classified as non-merger samples.

- (i) Projected physical separation (r_p): The projected distance to the closest galaxy in the pair must be $r_p < 100$ kpc.
- (ii) Redshift difference (Δz): The difference in redshift between the two galaxies must satisfy $\Delta z < 0.3$.
- (iii) Mass ratio (μ): The stellar mass of the lighter galaxy must be at least one-quarter of the stellar mass of the heavier galaxy, i.e. $\mu > 1/4$, for the pair to be classified as a major merger. Typically, galaxies with lower mass ratios ($1/10 < \mu < 1/4$) are classified as minor mergers. We present mass ratio distributions of our samples in Fig. A1.

The first criterion, projected physical separation, is widely used in previous studies of the local Universe with SDSS data (S. L. Ellison et al. 2008; D. R. Patton et al. 2011; J. M. Scudder et al. 2012; D. R. Patton et al. 2013), as well as at higher redshifts (E. A. Shah et al. 2022) ($0.5 < z < 3.0$). In our approach, we apply the K-Dimensional Tree (KD-Tree) algorithm (J. L. Bentley 1975), a space-partitioning data structure commonly used for nearest-neighbour searches, to efficiently compute the projected physical distance to the close neighbouring galaxies for each source in our sample. The mass ratio criterion for defining major mergers is also widely adopted in several studies (e.g. K. R. V. Casteels et al. 2014; C. López-Sanjuan et al. 2015; A. W. S. Man, A. W. Zirm & S. Toft 2016; C. J. Mundy et al. 2017; K. B. Mantha et al. 2018; K. Duncan et al. 2019; C. J. Conselice et al. 2022; Q. Duan et al. 2025; D. Puskás et al. 2025). The key difference in the classification criteria used in this work is the second one – the Redshift Difference (Δz) criterion.

Various works at lower redshift ($z < 3$) apply a rest-frame velocity difference criteria instead of a redshift difference. For example, in the local Universe, S. L. Ellison et al. (2008) apply a velocity difference limit of $\Delta v < 500$ km s⁻¹, while J. M. Scudder et al. (2012) apply a stricter limit of $\Delta v < 300$ km s⁻¹. At $0.5 < z < 3.0$, E. A. Shah et al. (2022) use both $\Delta v < 1000$ km s⁻¹ and $\Delta v < 5000$ km s⁻¹ as velocity difference limits. There are two main reasons for using Δz in this work. First, the previous papers all use spectroscopic observations, which enable highly accurate redshift measurements, whereas photometric redshifts are used in this work. Due to the faint nature of the high-redshift galaxies in our sample, spectroscopic observations require extended integration times of up to 28 hours. As a result, it is not possible to obtain spectroscopic observations for all galaxies detected photometrically. Additionally, the current number of available *JWST* spectroscopic observations are not statistically sufficient for this purpose. Secondly, the redshift range used in this work, $z = [4.5, 8.5]$, is wider than in all previous studies. Given the definition $\Delta v = \frac{c \times \Delta z}{(1+z_{\text{mean}})}$, the corresponding Δv value for a given Δz varies largely for pairs at different redshifts (e.g. a fixed $\Delta z = 0.3$ corresponds to $\sim 16,352$ km s⁻¹ at $z = 4.5$, but only $\sim 9,467$ km s⁻¹ at $z = 8.5$). Since the average photometric uncertainties in our sample are around 0.1, we choose $\Delta z < 0.3$ as our fiducial limit for classifying close-pair galaxies. In the next section, we present our results using our fiducial Δz criterion, and in Appendix B, we show and discuss results under five different Δz limits: 0.1, 0.2, 0.3, 0.4, and 0.5.

4 RESULTS

4.1 Merger-driven star formation activity

In this section, we describe the methods used to investigate the enhancement of star formation in galaxy pairs at various sep-

arations and present the corresponding results. We then compare our observational findings with predictions from the *Semi-analytic Forecasts for JWST* (SC-SAM) simulation, as detailed in Section 4.1.4.

Using the close-pair classifications described in Section 3, we classify galaxies into two groups. Those that satisfy all three close-pair criteria are placed in the close-pair group, while those that fail to meet any one of these criteria are classified as non-merger samples. The non-merger group is used as a control set of galaxies to investigate the extent of star formation enhancement in close-pair galaxies. We then separate our redshift range into two bins: $z = [4.5, 6.5]$ and $z = [6.5, 8.5]$, and the enhancement in each redshift bin is investigated separately. To maintain sample completeness, we use all galaxy samples with masses in the range $[8, 10] [\log_{10}(M_*/M_\odot)]$, as shown in Fig. 1. After doing this we then plot the SFR as a function of r_p and investigate the SFR enhancements at various separations. In total, the sample includes 420 galaxies with close companions at $z = [4.5, 6.5]$ and 76 at $z = [6.5, 8.5]$.

4.1.1 SFR enhancement as function of pair separation

In this section, we present the results of our investigation into SFR enhancement. Fig. 2 shows hexbin distributions of $\log(\text{SFR})$ at various projected separations, up to 100 kpc, for all major merger close-pair samples across two redshift bins: $z = [4.5, 6.5]$ and $z = [6.5, 8.5]$. The median SFR of the non-merger samples is represented by a dashed line, with the 16th and 84th percentile ranges displayed as a shaded region. Since we do not have a statistically large enough dataset for every 10 kpc bin, we subsequently binned the data with a bin width of 20 kpc and used the bootstrapping method (B. Efron 1979; B. Efron & C. Stein 1981) to calculate the SFR for each bin in order to maintain a sufficient number of sources in each bin. The results are illustrated in Fig. 3a. This figure also includes the SFRs of morphologically detected merging galaxies from the Gold Samples identified in N. Dalmasso et al. (2024), represented by gold symbols. The Gold Samples consist of galaxies whose morphology parameters satisfy the criteria from both C. J. Conselice et al. (2003) and J. M. Lotz et al. (2008) for classifying merging galaxies.

For $z = [4.5, 6.5]$, we detect enhancements in SFRs only at $r_p < 20$ kpc bin, with an increase of 0.25 ± 0.10 dex above the median of the non-merger samples, corresponding to a factor of 1.77 ± 0.35 . The SFR of close-pair samples at $r_p = 20$ -100 kpc is comparable to the median value of non-merger samples but is below the 84th percentile of non-merger samples. Similarly, for $z = [6.5, 8.5]$, the enhancements in SFRs are only seen at $r_p < 20$ kpc bin, with an increase of 0.26 ± 0.11 dex above the median of the non-merger samples, corresponding to a factor of 1.76 ± 0.49 . No enhancement signals are detected at projected separations of $r_p = 20$ -100 kpc. In particular, the SFR of close-pair samples at $r_p = [80, 100]$ kpc is approximately 0.36 dex lower than the median value of non-merger samples.

Since this is the first work to investigate SFR enhancement at the previously unexplored epoch of $z > 4.5$, we can only compare our findings with results at lower redshifts. E. A. Shah et al. (2022) used spectroscopic observations to investigate SFR enhancements at $0.5 < z < 3.0$, finding an enhancement factor of $\sim 1.23^{+0.08}_{-0.09}$ in the $r_p < 25$ kpc bin only. C. N. Lackner et al. (2014) identified merging galaxies in the COSMOS field between $0.25 < z < 1.00$ with $\log M_*/M_\odot > 10.6$, using an automated method

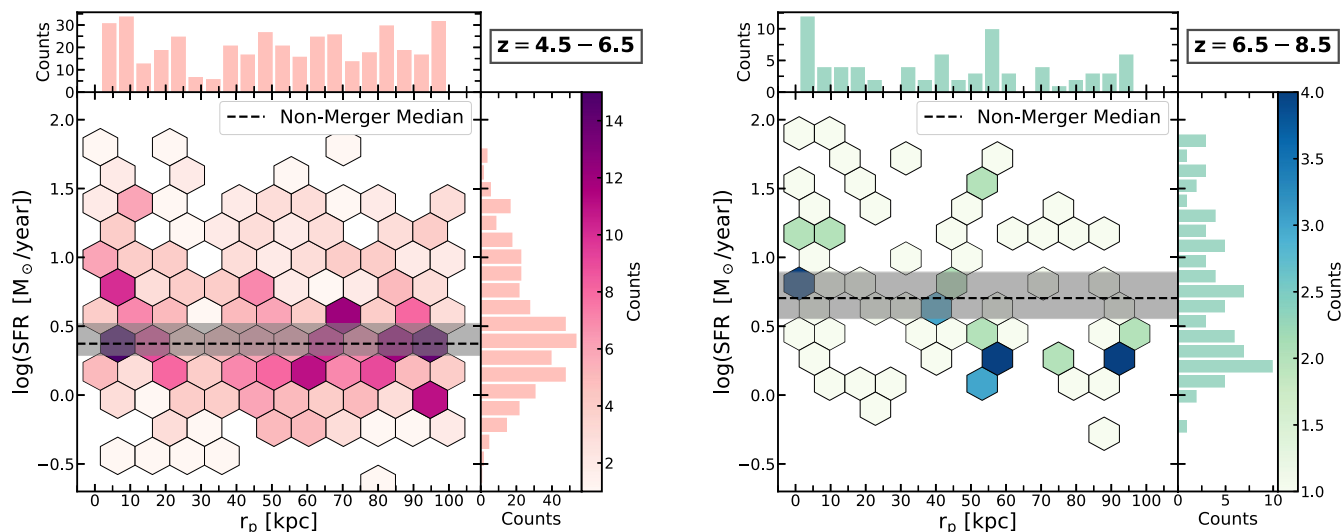


Figure 2. These two figures show the distribution of $\log(\text{SFR})$ as a function of projected separation to the closest neighbour in close-pair samples at $z = [4.5, 6.5]$ (Left) and $z = [6.5, 8.5]$ (Right). Close-pair samples are defined as those with $r_p < 100$ kpc, $\Delta z < 0.3$, and a mass ratio of $\mu > 1/4$. In each hexbin, darker colours indicate a higher number of galaxies in that bin. The histograms at the top and right of each figure show the distributions of r_p and $\log(\text{SFR})$ for the close-pair samples, respectively.

to median-filter high-resolution *HST* images from COSMOS to distinguish two concentrated galaxy nuclei at small separations (2.2–8 kpc). They reported an SFR enhancement of 2.1 ± 0.6 in their merging sample compared to their non-merging sample. At low redshifts, the projected separation range for enhancement is generally larger. For example, S. L. Ellison et al. (2008) found an enhancement range extending out to 30–40 kpc. J. M. Scudder et al. (2012) ($z < 1$) found statistically significant SFR enhancements extending to projected separations as far as 80 kpc. In contrast, A. Silva et al. (2018) found no significant star forming difference between merger and non-merger galaxies at $3 < r_p < 15$ kpc in the redshift range $0.3 < z < 2.5$, and T. J. Wilson et al. (2019) did not observe significant SFR enhancements in merging galaxy pairs compared to control samples of isolated galaxies at $1.5 \leq z \leq 3.5$.

High-redshift galaxies are generally more compact, with R_e averaging around 1 kpc, compared to 4–5 kpc at lower redshifts ($z = 1$ –2) (K. Ormerod et al. 2024). The smaller sizes of high-redshift galaxies naturally results in shorter permitted interaction ranges between close pairs. This size discrepancy may lead to a diminished scale of gravitational influence and less opportunity for gas inflow across extended distances. Furthermore, SFRs derived from alternative star formation histories (SFHs), including delayed, constant, and exponential models, also show enhancement. The degree of enhancement varies with the chosen SFH, exhibiting fluctuations of up to 30 per cent.

4.1.2 sSFR and stellar mass enhancement

Having found some SFR enhancements at $r_p < 20$ kpc bin in both redshift bins $z = [4.5, 6.5]$ and $z = [6.5, 8.5]$, we investigate two additional intrinsic galaxy properties linked with star formation which may vary with r_p : specific SFR (sSFR) and stellar mass. These properties could provide further insights into star forming and stellar population variations at different projected separations and redshifts. The reason for investigating stellar mass enhancement is that typical merger time-scales for close pairs

at $z = 4.5$ – 8.5 last around 50–100 Myr (S. Genel et al. 2014; M. Vogelsberger et al. 2014; G. F. Snyder et al. 2017; K. Duncan et al. 2019; C. J. Conselice et al. 2022; Q. Duan et al. 2025). It would be interesting to see whether the stellar mass increment from mergers during this period is significant or not. We apply the same method as in the SFR enhancement section (Section 4.1.1) to classify galaxies into close-pair and non-merger groups and investigate the variation of sSFR and stellar mass at different projected separations. The results are shown in Figs 3(b) and (c), respectively.

Fig. 3(b) shows the sSFR variations as a function of projected separation, r_p , for two redshift bins: $z = [4.5, 6.5]$ (left panel) and $z = [6.5, 8.5]$ (right panel). The red and blue diamonds represent the bootstrapped sSFR values for galaxies in close pairs across different bins, while the black dashed lines show the median sSFR for the non-merger sample. The shaded grey regions represent the 16th and 84th percentile ranges for the non-merger galaxies. Additionally, data points marked by gold diamonds correspond to the results from the N. Dalmasso et al. (2024) Gold Samples, which identify morphologically detected merging galaxies. The Gold Sample comprises galaxies whose morphological parameters meet the merger classification criteria defined by both C. J. Conselice et al. (2003) and J. M. Lotz et al. (2008).

In the $z = [4.5, 6.5]$ bin, we observe that the sSFR values for close-pair galaxies slightly exceed the non-merger median in the 20–60 kpc range, but remain within the 84th percentile range of the non-merger samples. No significant enhancement is detected at the smallest separation bin ($r_p < 20$ kpc). In the higher redshift bin, $z = [6.5, 8.5]$, the sSFR values show an enhancement at $r_p < 20$ kpc, with an increase of approximately 0.3 dex relative to the non-merger median, while at $r_p > 20$ kpc, the sSFR values are consistently lower than those of the non-merger samples. Overall, neither redshift bin demonstrates a strong or consistent sSFR enhancement at any separation, and close-pair galaxies do not show a clear trend of significant sSFR elevation relative to the non-merger sample at $z = [4.5, 8.5]$.

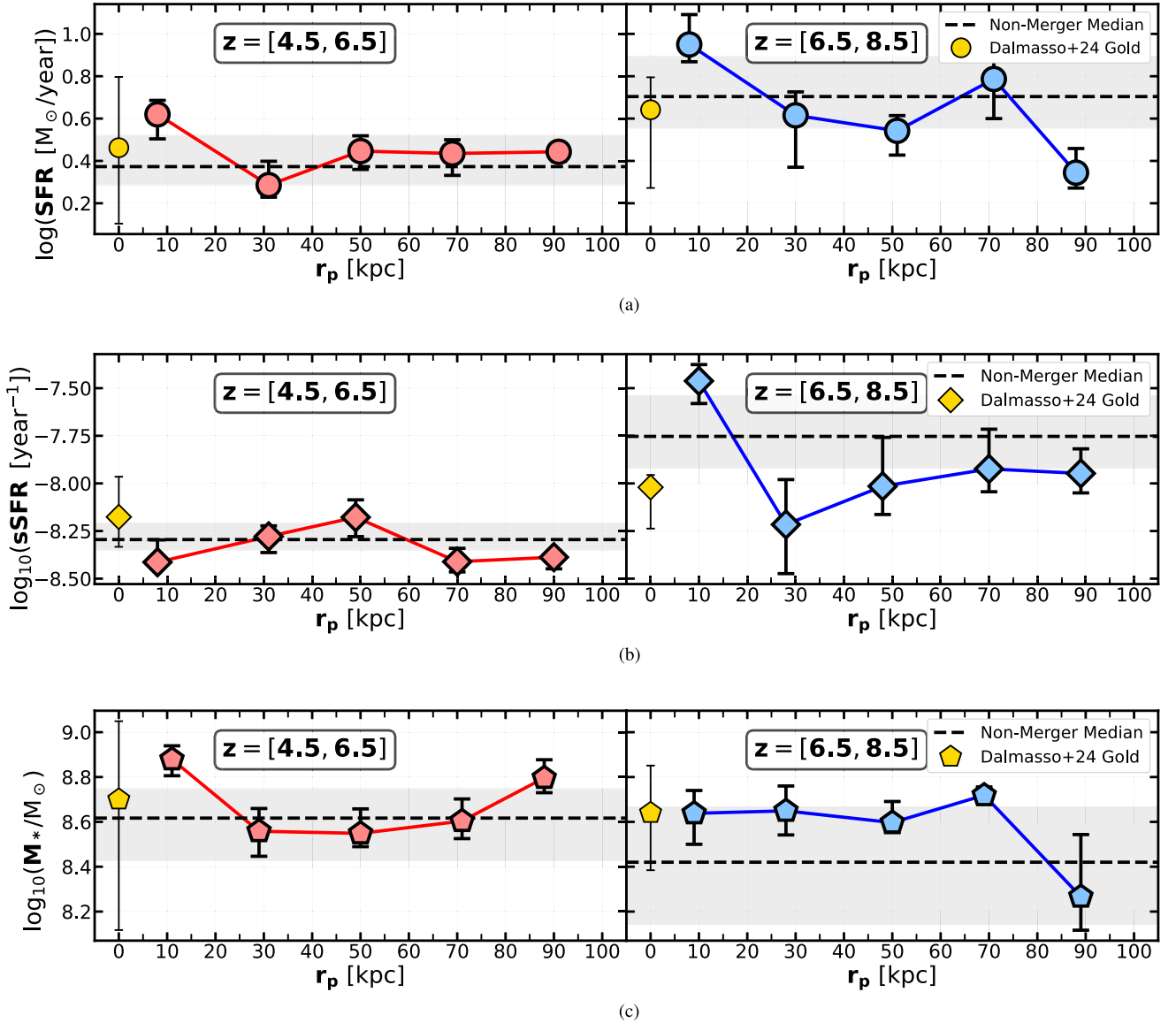


Figure 3. (a) The average value of the SFR (in log units) as a function of projected separation to the closest neighbour in close-pair samples at $z = [4.5, 6.5]$ and $z = [6.5, 8.5]$. At $z = [4.5, 6.5]$, we observe an enhancement of 0.25 ± 0.10 dex above the non-merger median at $r_p < 20$ kpc. Similarly, at $z = [6.5, 8.5]$, an enhancement of 0.26 ± 0.11 dex is observed at $r_p < 20$ kpc. No enhancement signals are detected at other r_p values in either redshift range. (b) Average sSFR as a function of projected separation to the closest neighbour in close-pair samples at $z = [4.5, 6.5]$ and $z = [6.5, 8.5]$. At $z = [4.5, 6.5]$, no enhancement in sSFR is observed across any projected separation range, with values oscillating around the non-merger median. At $z = [6.5, 8.5]$, an enhancement of ~ 0.3 dex above the non-merger median is detected at $r_p < 20$ kpc, while sSFR at $r_p > 20$ kpc shows values consistently below the non-merger median. (c) The average stellar mass for galaxies in pairs as a function of projected separation to the closest neighbour in close-pair samples at $z = [4.5, 6.5]$ and $z = [6.5, 8.5]$. At $z = [4.5, 6.5]$, an enhancement in stellar mass of 0.26 dex above the non-merger median is observed. At $z = [6.5, 8.5]$, the stellar mass of close-pair samples exceeds the non-merger median by 0.1 dex at $r_p < 80$ kpc. However, no strong evidence of significant stellar mass enhancement in close-pair galaxies compared to the non-merger sample is found. Three figures illustrate the evolution of galaxy SFR, sSFR, and stellar mass as a function of projected separation to the closest neighbour in close-pair samples computed from our observational data. The left panels present the results for $z = [4.5, 6.5]$, while the right panels show the results for $z = [6.5, 8.5]$. An enhancement in SFR is observed at $r_p < 20$ kpc. Results from morphology-identified merging samples from N. Dalmasso et al. (2024) are indicated by the yellow points at $r_p = 0$ kpc. Shaded regions represent the 16th and 84th percentile values.

In Fig. 3(c), we plot the variation of stellar mass as a function of projected separation, r_p , for two redshift bins: $z = [4.5, 6.5]$ (left panel) and $z = [6.5, 8.5]$ (right panel). In the $z = [4.5, 6.5]$ bin, we observe an enhancement in stellar mass at $r_p < 20$ kpc, with a value of approximately 0.26 dex above the non-merger median. At $r_p = 20$ – 80 kpc, the stellar mass of close-pair galaxies is comparable to the median value of non-merger galaxies, while

an additional enhancement of around 0.18 dex is observed at $r_p = 80$ – 100 kpc. For the $z = [6.5, 8.5]$ bin, the stellar mass of close-pair galaxies exceeds the median value of the non-merger sample at $r_p < 80$ kpc, though it remains within the 84th percentile of the non-merger distribution. These results suggest that while stellar mass enhancement is observed at $r_p < 20$ kpc for $z = [4.5, 6.5]$ and at $r_p < 80$ kpc for $z = [6.5, 8.5]$, the enhancement is not sta-

tistically significant, and the range of projected separations where it occurs varies considerably. Therefore, we conclude that there is no strong evidence for significant stellar mass enhancement in close-pair galaxies compared to the non-merger sample.

The above SFR, sSFR, and stellar mass evolution with projected separation to the closest neighbour are all based on a fiducial line-of-sight difference of $\Delta z < 0.3$. In Figs B1(a) and B1(b), we present results using various Δz upper limits, up to 0.5. Overall, we do not find noticeable differences between the results at different Δz upper limits.

4.1.3 Methodological challenges in comparing SFR offset between close-pair and controlled non-merger galaxy samples

A more statistically robust method for investigating SFR enhancements involves comparing the SFR offsets between close-pair and controlled non-merger isolated samples, effectively minimizing biases related to stellar mass, redshift, and environmental factors (e.g. J. Perez, P. Tissera & J. Blaizot 2009; S. L. Ellison et al. 2010; D. R. Patton et al. 2011; J. M. Scudder et al. 2012; L. E. Garduño et al. 2021; E. A. Shah et al. 2022). This approach requires constructing several controlled non-merger samples for each close-pair galaxy, matched closely in redshift and stellar mass. Low-redshift studies ($z < 1$) using SDSS data employ strict matching criteria when comparing galaxy samples. These studies (S. L. Ellison et al. 2010; D. R. Patton et al. 2011; J. M. Scudder et al. 2012) typically use a redshift difference limit of $\Delta z < 0.005$ and a stellar mass difference limit of $\Delta M_* < 0.1$ dex to ensure close matching. For each iteration, one additional control sample is paired with each merger sample, and a Kolmogorov–Smirnov (K–S) test is performed on the distributions of stellar mass and redshift to ensure similarity. If the p -value remains high, the process is repeated until the similarity criterion is no longer met. At redshifts ($0.5 < z < 3$), E. A. Shah et al. (2022) used the criteria of a stellar mass difference within 0.15 dex, a spectroscopic redshift difference within 0.15, and an overdensity difference within a factor of 1 to create controlled isolated samples.

Given the limited size of our data set at $z = [4.5, 6.5]$ (2404 galaxies) and $z = [6.5, 8.5]$ (926 galaxies), it is impractical to obtain a sufficient number of controlled non-merger isolated samples for each close-pair galaxy. We attempted this approach by applying broader matching criteria ($\Delta z < 0.1$ and $\Delta M_* < 0.1$ dex), but this still resulted in a rapid decline in the p -value of the K–S test after only a few iterations. Specifically, the p -value for stellar mass dropped to 0.41 after seven iterations, and for redshift, it decreased to 0.25 after four iterations. Therefore, while this method is valuable, its effectiveness relies on having a sufficiently large sample size, which could be achieved with future deep and wide-field NIRCам observations, larger than those from CEERS, JADES, and NEP-TDF.

4.1.4 Star formation trends from semi-analytic models

In this section, we investigate the evolution of SFR, sSFR, and stellar mass as a function of projected separation to the closest neighbour, using data from the *Semi-analytic Forecasts for JWST* (SC-SAM) simulation. The *Semi-analytic Forecasts for the Universe* project provides physically motivated predictions of galaxy and quasar properties and distributions across cosmic time, generated with the Santa Cruz semi-analytic model (SAM) for galaxy formation. This model strikes a balance between physical realism

and computational efficiency and has been shown to successfully reproduce observational constraints up to redshift $z \sim 10$ (L. Y. A. Yung et al. 2019a).

The project delivers two main data products: *Semi-analytic Forecasts for JWST* and *Semi-analytic Forecasts for Roman*. In this study, we utilize the *Semi-analytic Forecasts for JWST* data set. The methodology and details of the simulation are well-documented across several studies (L. Y. A. Yung et al. 2019a, b, 2020a, b, 2021, 2022).

The *deep* dataset comprises eight independent lightcone simulations. For each lightcone catalogue, we apply a consistent mass completeness limit of $8 < \log_{10}(M_*/M_\odot) < 10$ (The SC-SAM catalogue is complete at $\log_{10}(M_*/M_\odot) > 6.0$), which initially filters the catalogue and results in approximately 65 000 galaxies per lightcone. We then follow a methodology similar to that used for our observational data to classify galaxies into close-pair and non-merger samples. The only difference lies in the line-of-sight criterion: for observational data, we rely on Δz due to the use of photometric redshifts, whereas in the simulation, exact redshift values and the large sample size allow us to apply a more stringent selection based on rest-frame relative velocity between galaxies, Δv , with a fiducial upper limit of 1000 km s^{-1} . For reference, the observational criterion adopted in this work is $\Delta z < 0.3$ (Section 3), which corresponds to $\Delta v < 16,400 \text{ km s}^{-1}$ at $z = 4.5$ and $\Delta v < 9500 \text{ km s}^{-1}$ at $z = 8.5$.

The variations in SFR, sSFR, and stellar mass as a function of projected separation to the closest neighbour, using a line-of-sight velocity difference of $\Delta v < 1000 \text{ km s}^{-1}$, are shown in Figs 4(a)–(c). We first compute these properties as a function of projection separation for all eight realizations separately, using a bin width of 10 kpc and bootstrapping to obtain the median values. The results from all realizations are then combined, with the overall median calculated as the median of the eight realization medians. Similarly, the 16th and 84th percentiles are derived from the distribution of the medians across all realizations. The left panels display the results for $z = [4.5, 6.5]$, while the right panels correspond to $z = [6.5, 8.5]$. Additionally, results using various Δv upper limits, up to 5000 km s^{-1} , are shown in Figs B1(c) and (d). We also evaluated the results with Δv upper limits of 100, 200, and 300 km s^{-1} for the SC-SAM data, and found them to be closely aligned with the results from other Δv upper limits. As can be seen, we do not observe significant differences between the results across different Δv upper limits.

At $z = [4.5, 6.5]$, we observe a flat trend in SFR from $r_p = 100$ to 20 kpc , with values from the close-pair samples closely aligning with those of the non-merger samples. At $r_p < 20 \text{ kpc}$, there is a slight decrease of 0.1 dex below the non-merger median. However, this trend does not hold at $z = [6.5, 8.5]$, where we find close-pair samples' SFRs are $0.06^{+0.05}_{-0.02}$ dex above the non-merger median across nearly all projection separation bins. Although there are some fluctuations in the close-pair samples relative to the non-merger medians, these fluctuations are small, and we do not find any significant star formation enhancement at any projected separation and redshift bin.

For stellar mass at $z = [4.5, 6.5]$, we observe an opposite trend to the SFR variations in this redshift range. The stellar mass oscillates around the non-merger median from $r_p = 100 \text{ kpc}$ to $r_p = 10 \text{ kpc}$, with amplitude of approximately 0.05 dex. At $r_p < 10 \text{ kpc}$, an increase of 0.1 dex in stellar mass is observed. The stellar mass evolution at $z = [6.5, 8.5]$ follows a similar trend to the SFR evolution in this redshift range.

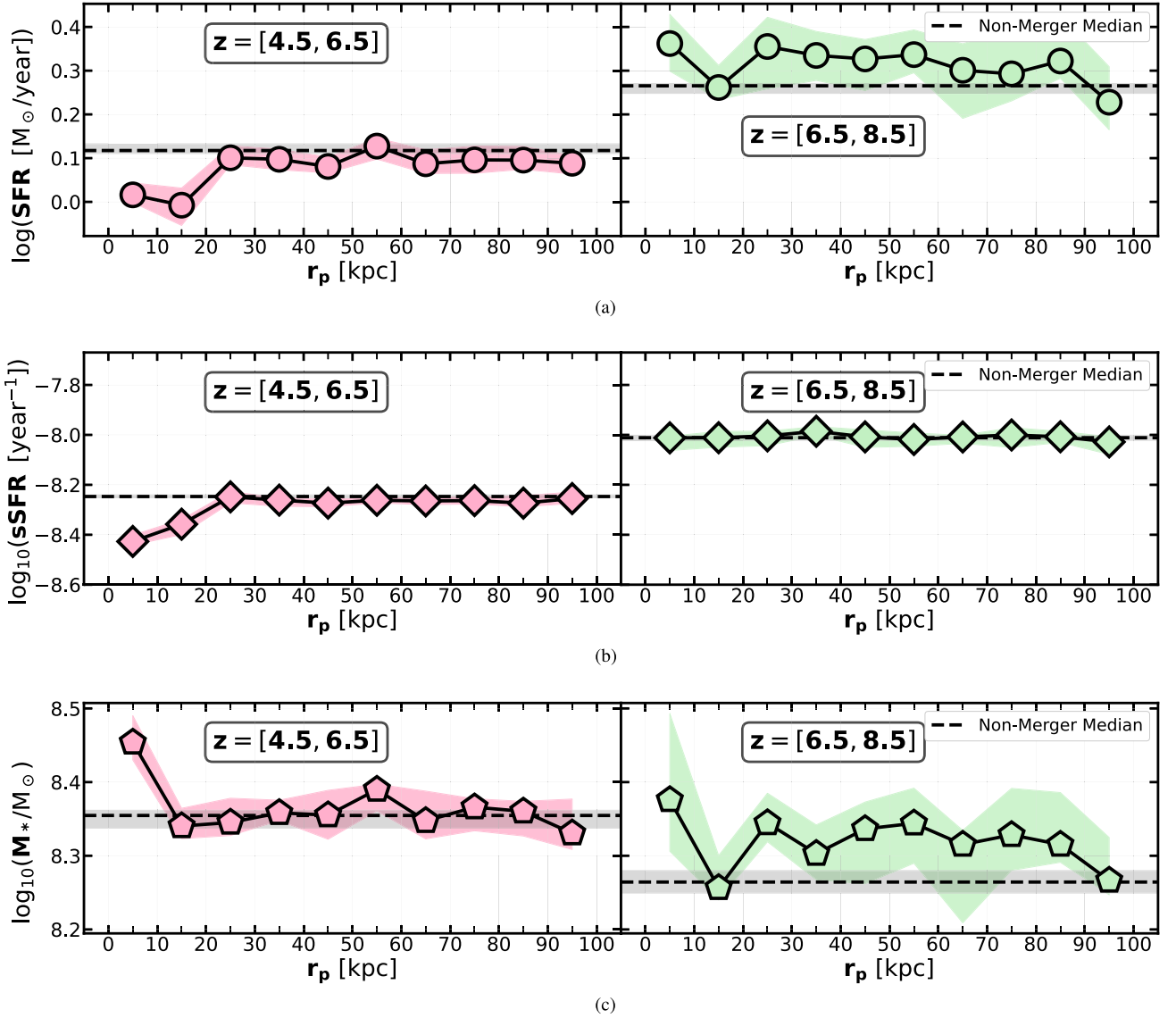


Figure 4. (a) Averaged SFRs as a function of projected separation to the closest neighbour in close-pair samples at $z = [4.5, 6.5]$ and $z = [6.5, 8.5]$. No SFR enhancement signals are detected at any projected separation. (b) Averaged sSFRs as a function of projected separation to the closest neighbour in close-pair samples at $z = [4.5, 6.5]$ and $z = [6.5, 8.5]$. No enhancement signals are detected at any projected separation. (c) Average stellar mass as a function of projected separation to the closest neighbour in close-pair samples at $z = [4.5, 6.5]$ and $z = [6.5, 8.5]$. No strong evidence of significant stellar mass enhancement is observed. Figures showing the evolution of galaxy SFR, sSFR, and stellar mass as a function of projected separation to the closest neighbour in close-pair samples with $\Delta v < 1000 \text{ km s}^{-1}$ computed from the SC-SAM simulation *deep* dataset. The left panels presents the results for redshifts $z = [4.5, 6.5]$, while the right panels show the results for $z = [6.5, 8.5]$. We first compute these properties with projection separation for all eight realizations in the *deep* catalogue separately, using a bin width of 10 kpc and bootstrapping to obtain the median values. The results from all realizations are then combined, with the overall median calculated as the median of the eight realization medians. The 16th and 84th percentiles are derived from the distribution of the medians across all realizations, and are shown by the shaded regions. Overall, we do not find any star formation enhancements at any projected separation.

Since sSFR is defined as the ratio of SFR to stellar mass, the observed trends at $z = [4.5, 6.5]$ – a decrease in SFR at $r_p < 20$ kpc and an increase in stellar mass at $r_p > 10$ kpc – result in a decrease in sSFR at $r_p < 20$ kpc. Between $r_p = 20$ and 100 kpc, the sSFR of the close-pair samples closely aligns with the non-merger medians. At $z = [6.5, 8.5]$, sSFR remains close to the non-merger medians at all projection separations.

Overall, while we observe some variations in SFR, sSFR, and stellar mass in the close-pair samples relative to the non-merger

samples at $r_p < 20$ kpc from SC-SAM simulation, these variations are not significant and do not indicate any star formation enhancement at any projected separation. This suggests that the gas inflow resulting from mergers may be underestimated in the simulation, and thus does not sufficiently trigger star formation.

While close-pair interaction-induced star formation may not be explicitly included in the semi-analytic model, star formation rate (SFR) enhancement following mergers (i.e. post-merger phases) is taken into account. In Fig. 5, we show the SFR as a function of

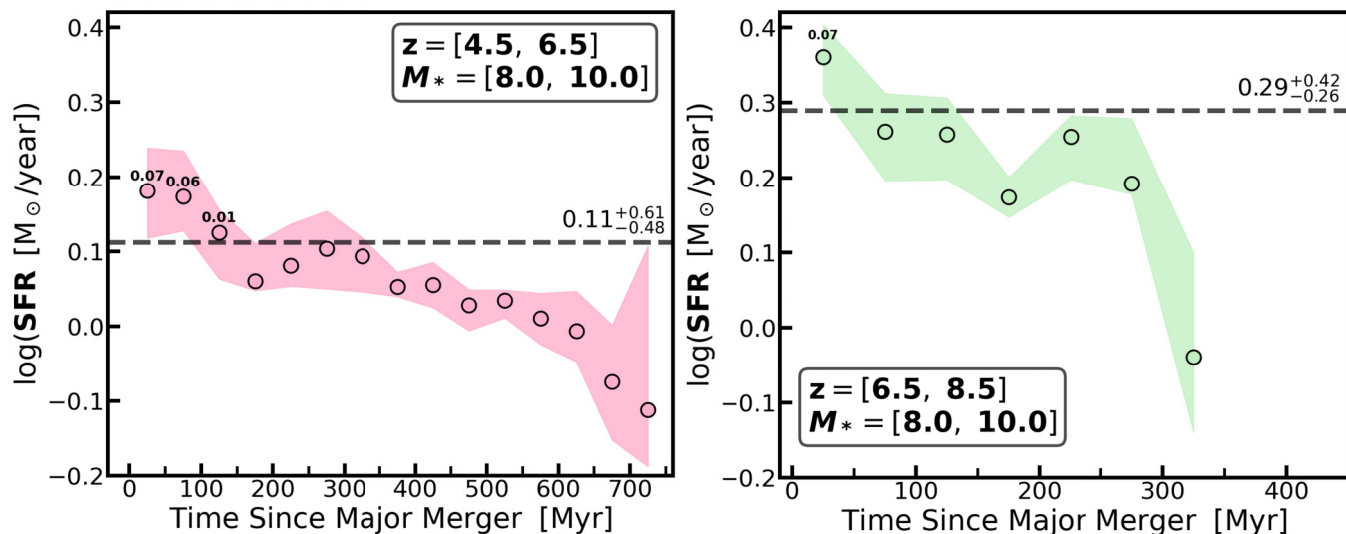


Figure 5. SFR enhancement as a function of time since the last major merger, shown for two redshift bins: $z = [4.5, 6.5]$ (left) and $z = [6.5, 8.5]$ (right). In each panel, the median SFR of the non-merger sample is indicated by a horizontal dashed line, with the 16th and 84th percentile values labelled at the top right of each line. The level of SFR enhancement relative to the non-merger median is annotated above each data point. Results are computed across 8 SC-SAM realizations, with each data point showing the median across realizations, and the 16th–84th percentile range indicated by the shaded regions. Overall, SFR is most strongly enhanced immediately after the merger, followed by a steady decline over time, eventually leading to quenching.

time since the last major merger. The left column presents results for the redshift range $z = [4.5, 6.5]$, while the right column corresponds to $z = [6.5, 8.5]$. For reference, the median SFR of non-merging galaxies in each mass and redshift bin is indicated by a horizontal dashed line, with the 16th and 84th percentiles shown as error bands. Above each data point, we annotate the level of SFR enhancement relative to the non-merger median. These enhancements are computed across all 8 SC-SAM realizations, with the median values plotted and the corresponding 16th–84th percentile spread shown as shaded regions. We find that SFR enhancement peaks immediately after the merger and steadily declines over time, eventually leading to quenching. This post-merger quenching behaviour is consistent with observational results from S. L. Ellison et al. (2022) and L. Ferreira et al. (2024). From the SC-SAM results, it appears that detecting enhanced SFR requires catching galaxies shortly after the merger, as the SFR drops to the level of non-merging systems within 100–200 Myr.

4.2 Mergers trigger AGN activity

The unexpectedly high population of AGNs and gravitational wave detections in the early Universe remains a mystery (e.g. B. P. Abbott et al. 2016, 2019; J. Scholtz et al. 2023; I. Juodžbalis et al. 2025; R. Maiolino et al. 2025). High-redshift galaxies alone are not expected to form AGNs within such short time-scales at these early epochs. One possible explanation is the frequent occurrence of galaxy mergers. Galaxy merger rates at $z < 6$ have been extensively investigated (e.g. D. R. Patton et al. 2002; C. J. Conselice 2003; C. R. Bridge et al. 2007; C. J. Conselice, S. Rajgor & R. Myers 2008; C. López-Sanjuan et al. 2015; A. W. S. Man et al. 2016; C. J. Mundy et al. 2017; E. Ventou et al. 2017; K. B. Mantha et al. 2018; K. Duncan et al. 2019; C. J. Conselice et al. 2022), though the galaxy merger rate at very high redshift ($z > 6$) had not been studied in detail until the recent observational study by Q. Duan et al. (2025) and D. Puskás et al. (2025). They presented

pair fractions, merger rates, and stellar mass accretion rates at $z > 6$, revealing rapid galaxy merger rates of approximately 6 mergers per Gyr. In this section, building on their results, we further investigate the AGN fractions and AGN excess in close-pair samples. As in addition to SFR, it is possible that close pairs of galaxies and mergers will induce AGN activity in galaxies.

4.2.1 AGN identification

In addition to the photometric data, we incorporate public *JWST*/NIRSpec spectroscopic data to identify Type-I and Type-II AGNs. While the previous sections on merger-triggered star formation utilized data from all eight fields (CEERS, JADES GOODS-S, NEP-TDF, NGDEEP, GLASS, El-Gordo, SMACS-0723, and MACS-0416), only the CEERS and JADES GOODS-S fields contain a substantial number of spectroscopic observations. Therefore, this section focuses on measuring AGN fractions and AGN excess in these two fields.

We employ two independent methods to identify AGNs: the photometric SED method (Type-I) (e.g. R. J. Assef et al. 2010; L. Ciesla et al. 2015; I. Juodžbalis et al. 2023) and the spectroscopic BPT (Baldwin–Phillips–Terlevich) diagnostic method (Type-II) (S. Veilleux & D. E. Osterbrock 1987; L. J. Kewley et al. 2001; G. Kauffmann et al. 2003; L. J. Kewley et al. 2006). We discuss the BPT method in detail below and present results from both approaches. The photometric SED method, along with the relevant AGN templates, is outlined in I. Juodžbalis et al. (2023). In brief, this approach is based on SED fitting using *JWST*/NIRCam and MIRI photometry. It models both galaxy and AGN components with X-CIGALE, identifying AGNs through mid-infrared excess emission characteristic of hot dust heated by AGN activity.

For the BPT method, AGN identification becomes more challenging at high redshift ($z > 3$) due to the rapid evolution towards lower metallicities and higher ionization parameters. Neverthe-

less, several studies have shown that AGN detection via BPT diagnostics remains feasible under these conditions (e.g. M. Perna et al. 2023; J. Scholtz et al. 2023; G. Mazzolari et al. 2024).

Out of the 2525 PRISM spectra in the JADES GOODS-S catalogue, we use all spectra with spectroscopic redshifts categorized as A, B, or C. These categories are defined by the presence of at least two emission lines or a combination of a line and a strong continuum break in the PRISM spectra (A. J. Bunker et al. 2023; F. D'Eugenio et al. 2024). After applying this criterion, 1333 galaxies remain. We then apply the BPT diagnostic diagram to identify AGN based on line ratios of $H\beta$ $\lambda 4861/[O\ III]\lambda 5007$ and $[N\ II]\lambda 6583/H\alpha$. The resolution of the PRISM spectra ($R \sim 100$) is not sufficient to resolve $[N\ II]\lambda 6583$ from $H\alpha$. Therefore, we estimate the $[N\ II]\lambda 6583$ line flux by first deriving the $H\alpha$ flux from the observed $H\beta$ flux, assuming a dust-free Balmer line ratio of 2.86, and then subtracting this value from the combined $H\alpha + [N\ II]$ flux. While this method is traditionally applied at lower redshifts, recent studies (e.g. M. Perna et al. 2023; G. Mazzolari et al. 2024) have shown that it remains applicable at $z > 4$ under the assumption of Case B recombination and low dust content, which is consistent with expectations for many galaxies in the early Universe. Although uncertainties increase due to potential variations in dust attenuation and ionization conditions, this approach provides a reasonable first-order estimate of the $[N\ II]/H\alpha$ ratio for use in BPT diagnostics at high redshift.

Given the wavelength coverage of NIRSpec PRISM spectra (6000–53 000 Å), we can simultaneously detect the aforementioned AGN diagnostic lines only within the redshift range of $5.3 < z < 6.7$. In total, we identify 40 AGNs in GOODS-S, of which 19 are successfully matched with the EPOCHS photometric catalogue using a matching radius of < 0.32 arcsec. The choice of 0.32 arcsec is dictated by our source-extraction procedure: each object's photometry is measured in a circular aperture with radius 0.16 arcsec (i.e. a diameter of 0.32 arcsec, Section 2). Adopting the same diameter for the positional cross-match guarantees internal consistency between the astrometry and photometry and minimises the risk of spurious associations.

Additionally, by cross-matching the EPOCHS photometric catalogue with BPT-confirmed AGNs from the CEERS PRISM spectra catalogue produced by DAWN, we identify and match 25 AGNs. The positions of these 44 AGNs in the BPT diagram are marked with pink pentagons (for JADES) and green circles (for CEERS) in Fig. 6.

4.2.2 AGN fraction of galaxies in pairs

We follow criteria (i) and (ii) outlined in Section 3 to classify galaxies into close-pair and non-merger samples. Close-pair galaxies are defined as those with $\Delta z < 0.3$ and $r_p < 100$ kpc. The reason for not applying criterion (iii) – the major merger criterion – is that our goal is to explore all possible AGN presence in pairs, regardless of the merger mass ratio (Section 3, Fig. A1). We also do not have a large enough sample to investigate differences in mass ratios and separations. We thus are searching for a detection of significance, rather than one of measurement given our limited data. Furthermore, while the first part of this paper on merger-induced star formation enhancement investigates the redshift range $4.5 < z < 8.5$, we are only able to focus on AGN excess in the narrower redshift range of $5.3 < z < 6.7$ for merger-triggered AGN activity. This is because the NIRSpec wavelength coverage

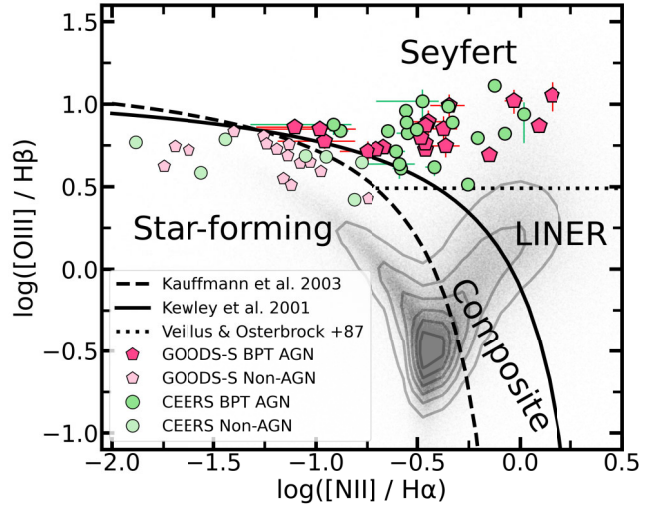


Figure 6. BPT diagnostic diagram, based on criteria from S. Veilleux & D. E. Osterbrock (1987); L. J. Kewley et al. (2001); G. Kauffmann et al. (2003); L. J. Kewley et al. (2006). Galaxies classified as AGNs are shown as red pentagons for the JADES GOODS-S field and green circles for the CEERS field, while non-AGNs are labelled with pink pentagons for JADES GOODS-S and light-green circles for CEERS. The contours are derived from 927 552 low-redshift SDSS DR7 galaxies, as documented by K. N. Abazajian et al. (2009).

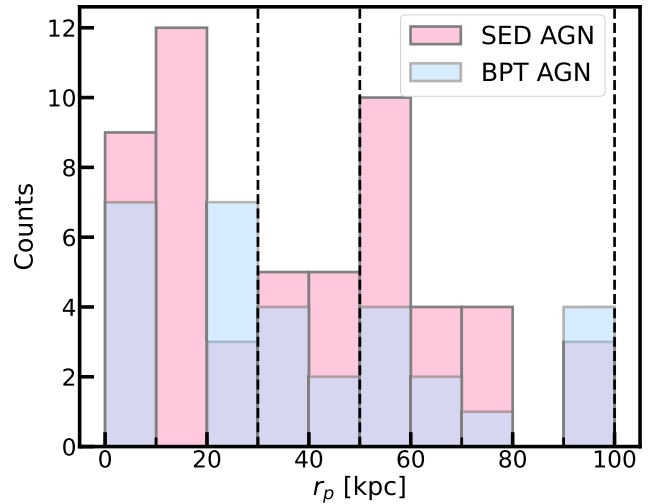


Figure 7. Distribution of projected separations between AGNs and their closest paired galaxy, measured out to a maximum of 100 kpc. The red histogram represents AGNs identified via SED (Type-I) fitting (I. Juodžbalis et al. 2023), while the blue histogram corresponds to AGNs identified using the BPT diagnostic diagram (Type-IIa) (S. Veilleux & D. E. Osterbrock 1987; L. J. Kewley et al. 2001; G. Kauffmann et al. 2003; L. J. Kewley et al. 2006). The three dashed black lines indicate projected separations of $r_p = 30, 50$, and 100 kpc, respectively. For SED-based AGNs, 24 AGN–galaxy pairs have separations less than 30 kpc, 10 fall within $30 < r_p < 50$ kpc, and 21 lie in the range $50 < r_p < 100$ kpc. For BPT-based AGNs, the corresponding numbers are 14, 6, and 11, respectively.

only allows simultaneous detection of the BPT diagnostic lines within this redshift range (see Section 4.2.1). In Fig. 7, we present two histograms showing the distribution of projected separations between AGNs and their closest companion galaxies. The red

histogram represents Type-I AGNs identified through SED fitting, while the blue histogram corresponds to Type-II AGNs identified using the BPT diagnostic diagram. The 16th, 50th, and 84th percentiles of r_p to the closest companion are [9.8, 36.2, 64.2] kpc for SED-selected Type-I AGNs and [8.5, 37.9, 69.9] kpc for BPT-selected Type-II AGNs, respectively.

We present three distinct definitions of AGN fractions. The specific quantities measured are defined as follows:

(i) $f_{\text{galaxy-pair,AGN}}$: the fraction of AGNs in pairs relative to the total number of AGNs, calculated by dividing the number of AGNs in pairs by the total number of AGNs. For example, if this value is $= 1$ this would imply that all AGN are in a galaxy pair, likewise if $= 0$ no AGN are found in any galaxies in pairs.

(ii) $f_{\text{AGN,close-pair}}$: the fraction of AGNs in close-pair merger samples, calculated by dividing the number of AGNs in pairs by the total number of galaxies in pairs. A value $= 1$ would mean that all galaxies in pairs have an AGN, and $= 0$ would imply that no galaxies in pairs contain an AGN.

(iii) $f_{\text{AGN,isolated}}$: the fraction of AGNs in non-merger samples, calculated by dividing the number of AGNs not in pairs by the total number of galaxies not in pairs. Similar to above, a value $= 1$ would imply all isolated galaxies have an AGN and $= 0$ would imply no isolated galaxies have an AGN.

We present these fractions $f_{\text{galaxy-pair,AGN}}$, $f_{\text{AGN,close-pair}}$, and $f_{\text{AGN,isolated}}$, alongside the redshift evolution of galaxy pair fractions modelled using both the power-law and power-law + exponential models from Q. Duan et al. (2025), in Figs 8(a)–(c), within the redshift range $5.3 < z < 6.7$. Numerical values of each AGN Fraction is shown in Table 1. The galaxy pair fractions from Q. Duan et al. (2025) are defined as the number of galaxy pairs relative to the total number of galaxies, and are computed using a statistical probability pair-counting method. The left panel of each figure shows the Type-II AGN fraction for AGNs identified using the BPT diagram, while the right panel shows the AGN fraction for Type-I AGNs identified using the photometric SED method. AGN fractions from JADES GOODS-S are represented by circles, while those from CEERS are indicated by crosses. The colours correspond to the maximum projected separation between pairs: green, red, and yellow represent the maximum separations of 30, 50, and 100 kpc, respectively. The median value of all data points in each AGN fraction figure is represented by a dotted line, with the 16th and 84th percentile values serving as the 1σ confidence intervals. Given the small number statistics, where the distributions deviate from a perfect Poisson distribution, the uncertainties on the AGN fractions are computed following the method described by N. Gehrels (1986) for small number counts. Specifically, equations (9) and (14) are used to determine the upper and lower confidence limits, adopting $S = 1.645$, $\beta = 0.031$, and $\gamma = -2.50$, which correspond to a 95 per cent confidence level under the assumption of a Poisson distribution. Comparing AGN fractions derived from the BPT and SED methods, we find that for a same AGN fraction definition, the results from SED-based and BPT-based AGNs are generally in good agreement – except for $f_{\text{AGN,close-pair}}$, defined as the number of AGNs in pairs relative to the total number of galaxies in pairs – where the AGN fraction from SED-based AGNs ($0.043^{+0.010}_{-0.012}$) is approximately 1.8 times higher than that from BPT-based AGNs ($0.024^{+0.006}_{-0.006}$). This discrepancy is expected, as the AGNs identified by the SED and BPT methods represent two distinct populations: Type-I and Type-II AGNs, respectively. We will discuss this distinction in more detail in the Discussion section. Also, it is worth noting that

the results from the CEERS field are consistently lower than those from JADES, which can be attributed to cosmic variance (B. P. Moster et al. 2011) and the more uncertain redshift estimates in CEERS due to the lack of F090W band coverage.

Furthermore, A direct result that can be drawn from Fig. 8(a) is that half of the AGNs are found in pairs at $r_p < 50$ kpc and $\Delta z < 0.3$. This fraction ($f_{\text{galaxy-pair,AGN}}$) is $3.25^{+1.50}_{-1.06}$ times higher the galaxy pair fractions, even at the smallest projection separation limit of $r_p < 30$ kpc. The distribution of $r_p < 100$ kpc points shows that almost all AGNs have a companion within 100 kpc in projection and within $\Delta z < 0.3$ along the line of sight. This suggests that galaxy interactions, as indicated by the presence of a close companion, play a crucial role in facilitating or enhancing conditions favorable for triggering AGN activity. It is important to emphasize that in this paper we investigate AGN fractions in close-pair samples, rather than in galaxies that are confirmed to be merging or in post-merger phases. Therefore, our results do not necessarily imply that physical merger events directly trigger AGN activity.

Previous studies at low redshift using SDSS data, such as S. L. Ellison et al. (2011); S. Satyapal et al. (2014), show that $f_{\text{AGN,close-pair}}$ increases with decreasing r_p , rising by a factor of up to ~ 2.5 at $r_p < 10$ kpc. However, we do not observe such a trend in our data due to the limited sample size, which hinders our ability to trace this projected separation evolution in detail.

Despite the differences in AGN fractions between the two methods, the key parameter in this section is the AGN excess (F. Gao et al. 2020; J. M. Comerford et al. 2024; E. Koulouridis, A. Gkini & E. Drigga 2024), defined as the ratio $f_{\text{AGN,close-pair}}/f_{\text{AGN,isolated}}$ (Fig. 8b/Fig. 8c). We call this the ‘AGN Excess’ which is in many ways a better quantity to determine than the individual AGN fractions. The reason for this is that the methods for locating and identify AGN is bound to have errors, biases, incompleteness and contamination. To first order, we can assume that these biases are the same for those in pairs as compared to those which are isolated. The AGN excess from the BPT method is $1.26^{+0.15}_{-0.04}$, while the SED method shows a value of $1.34^{+0.23}_{-0.11}$. Although the AGN excess values are not highly significant and come with considerable uncertainties, the consistent detection of AGN excess from two independent methods supports the robustness of this result of a slight, but significant excess.

Compared to other studies on AGN excess, the work by F. Gao et al. (2020), which utilized the SDSS and GAMA surveys to study AGN fractions up to $z = 0.6$, found an AGN excess up to 1.81 ± 0.16 . Using MaNGA data, J. M. Comerford et al. (2024) found an AGN excess of ~ 1.8 in major mergers and 1.5 in minor mergers.

These studies used large sample sizes to compute the AGN excess. In our case, given the limitations in the number of samples, the AGN excess found in our work is likely a lower limit, with the actual AGN excess potentially similar or higher than those reported in aforementioned studies at low redshift.

5 DISCUSSION

While this work represents an early attempt to investigate the impact of mergers on galaxy star formation and AGN properties at $z > 4.5$, several aspects and limitations should be carefully considered.

First, this study is based on photometric redshifts. As a result, we adopt a redshift difference for the line-of-sight criterion rather than a stricter velocity difference. Without precise spectroscopic redshifts, some of the close-pair systems identified

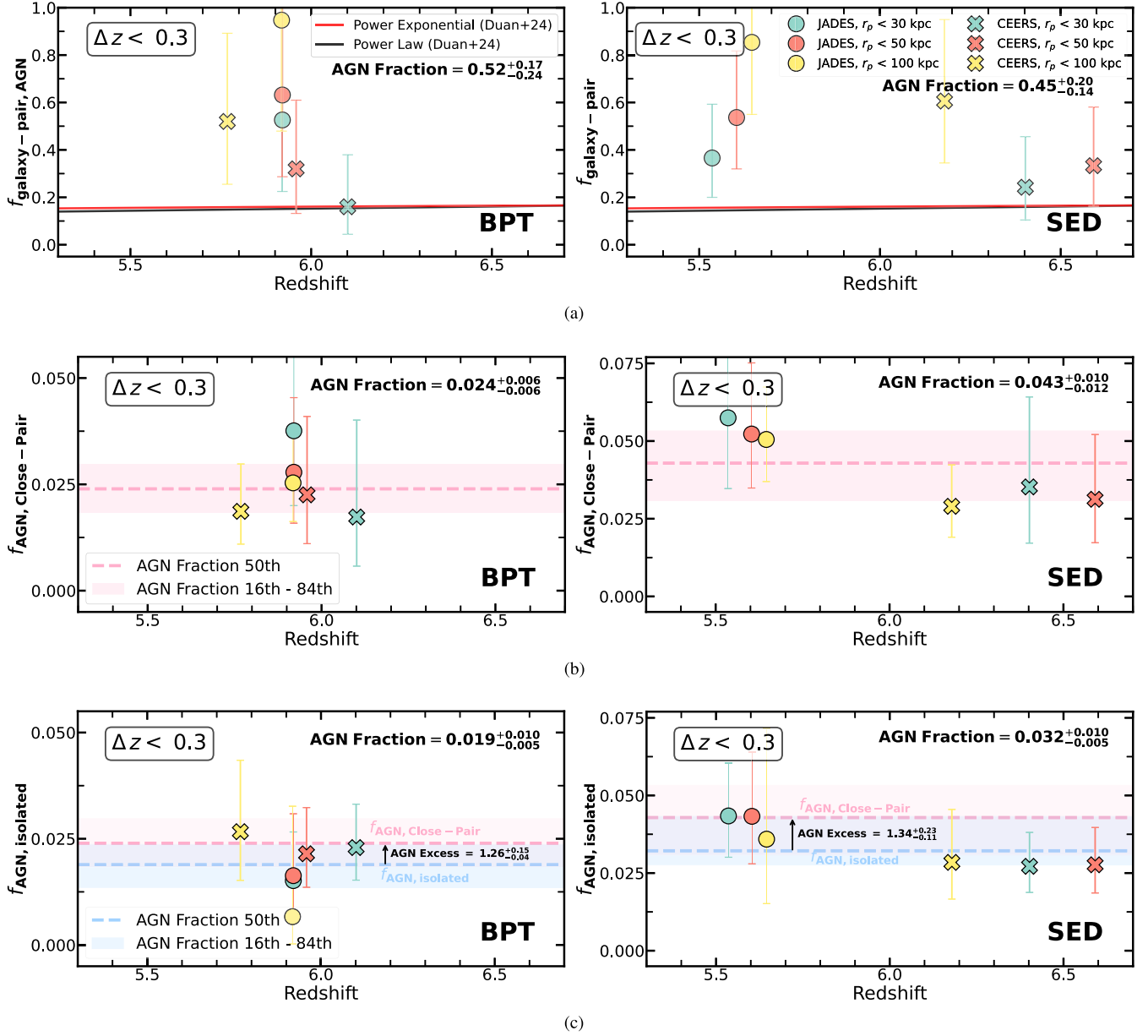


Figure 8. (a) Fraction of AGNs in pairs relative to the total number of AGNs, calculated by dividing the number of AGNs in pairs by the total number of AGNs. Circles represent JADES AGNs and crosses denote CEERS AGNs. AGN fractions calculated with projected separation upper limits of 30, 50, and 100 kpc are coloured in green, red, and yellow, respectively. We find that the fraction of AGNs in pairs is $3.25^{+1.50}_{-1.06}$ times higher than the fraction of galaxies in pairs, with almost all AGNs ($73^{+27}_{-32}\%$) having a companion within 100 kpc. (b) The fraction of AGNs in close-pair merger samples, calculated by dividing the number of AGNs in pairs by the total number of galaxies in pairs. Circles indicate JADES AGNs, while crosses represent CEERS AGNs. AGN fractions calculated with projected separation upper limits of 30, 50, and 100 kpc are coloured in green, red, and yellow, respectively. Red dashed line and shaded regions indicate the 50th, 16th, and 84th percentile values. (c) The fraction of AGNs in non-merger samples, calculated by dividing the number of AGNs not in pairs by the total number of galaxies not in pairs. Circles indicate pairs from the JADES data set and crosses represent pairs from the CEERS data set. AGN fractions calculated with projected separation upper limits of 30, 50, and 100 kpc are coloured in green, red, and yellow, respectively. Blue dashed line and shaded regions indicate the 50th, 16th, and 84th percentile values. Three figures showing three different AGN fractions: $f_{\text{galaxy-pair, AGN}}$, $f_{\text{AGN, close-pair}}$, and $f_{\text{AGN, isolated}}$ in our samples. Results for AGNs identified using the BPT (Type-II) diagnostic diagram are presented in the left figures, while results for AGNs selected using photometric SED (Type-I) are shown in the right figure. JADES GOODS-S data points are shown as circles, and CEERS points are shown as crosses. For each figure, the median AGN fractions are indicated by dotted lines, with shaded regions representing the 16th and 84th percentile values. Values of each AGN fraction are presented in Table 1. Galaxy pair fractions from the power-law model and the power-law + exponential model from (Q. Duan et al. 2025) are shown as black and red lines. An AGN excess of $1.26^{+0.15}_{-0.04}$ and $1.34^{+0.23}_{-0.11}$ is detected using the BPT and SED methods, respectively. We do not observe an increase in AGN fraction as close-pair separation decreases. Please note again that for the AGN-galaxy fraction, we do not apply any mass ratio constraint; i.e. both major and minor mergers are included.

Table 1. Two Tables presenting the three AGN fractions measured in this work: $f_{\text{galaxy-pair,AGN}}$, $f_{\text{AGN, close-pair}}$, and $f_{\text{AGN, isolated}}$, as defined in Section 4.2.2, for AGNs identified via the BPT (Type-II) and SED (Type-I) methods. The criteria used to define whether an AGN has a companion are described in Section 3, with results shown for projected separation thresholds of $r_p < 30$, 50, and 100 kpc. The top table presents the computed values of each AGN fraction, and the bottom table presents the AGN or galaxy counts used in the numerators and denominators for calculating the three AGN fractions.

Method	r_p (kpc)	$f_{\text{galaxy-pair,AGN}}$		$f_{\text{AGN, close-pair}}$		$f_{\text{AGN, isolated}}$	
		JADES	CEERS	JADES	CEERS	JADES	CEERS
BPT	30	$0.53^{+0.44}_{-0.30}$	$0.16^{+0.22}_{-0.12}$	$0.038^{+0.027}_{-0.018}$	$0.017^{+0.023}_{-0.012}$	$0.015^{+0.011}_{-0.007}$	$0.023^{+0.010}_{-0.008}$
	50	$0.63^{+0.39}_{-0.34}$	$0.32^{+0.29}_{-0.19}$	$0.028^{+0.018}_{-0.012}$	$0.023^{+0.018}_{-0.011}$	$0.016^{+0.015}_{-0.009}$	$0.021^{+0.011}_{-0.008}$
	100	$0.95^{+0.04}_{-0.47}$	$0.52^{+0.37}_{-0.27}$	$0.025^{+0.012}_{-0.009}$	$0.019^{+0.011}_{-0.008}$	$0.007^{+0.026}_{-0.006}$	$0.027^{+0.017}_{-0.011}$
SED	30	$0.37^{+0.23}_{-0.17}$	$0.24^{+0.21}_{-0.14}$	$0.057^{+0.032}_{-0.023}$	$0.035^{+0.029}_{-0.018}$	$0.043^{+0.017}_{-0.013}$	$0.027^{+0.011}_{-0.008}$
	50	$0.54^{+0.28}_{-0.22}$	$0.33^{+0.25}_{-0.17}$	$0.052^{+0.023}_{-0.017}$	$0.031^{+0.021}_{-0.014}$	$0.043^{+0.021}_{-0.015}$	$0.028^{+0.012}_{-0.009}$
	100	$0.85^{+0.15}_{-0.30}$	$0.61^{+0.34}_{-0.26}$	$0.051^{+0.017}_{-0.014}$	$0.029^{+0.013}_{-0.010}$	$0.036^{+0.036}_{-0.021}$	$0.028^{+0.017}_{-0.012}$
Method	r_p (kpc)	$f_{\text{galaxy-pair,AGN}}$		$f_{\text{AGN, close-pair}}$		$f_{\text{AGN, isolated}}$	
		JADES	CEERS	JADES	CEERS	JADES	CEERS
BPT	30	10/19	4/25	10/266	4/231	9/594	21/916
	50	12/19	8/25	12/431	8/355	7/429	17/792
	100	18/19	13/25	18/710	13/697	1/150	12/450
SED	30	15/41	8/33	15/261	8/227	26/599	25/920
	50	22/41	11/33	22/421	11/352	19/439	22/795
	100	35/41	20/33	35/693	20/690	6/167	13/457

may not actually be physically close and may have significant redshift differences. Although we utilized an extensive datasets available to date (CEERS, JADES GOODS-S, NEP-TDF, NGDEEP, GLASS, El-Gordo, SMACS-0723, and MACS-0416), the sample size is still limited, and the lack of spectroscopic redshifts is a more prominent drawback. Based on our observational results on SFR, sSFR, and stellar mass variations (Figs 3b–c), apart from the SFR enhancement at $r_p < 20$ kpc observed in both redshift bins $z = [4.5, 6.5]$ and $z = [6.5, 8.5]$, the other results do not show statistically significant evolutionary trends.

Second, the SFR, sSFR, and stellar mass values in this work are derived from photometric SED fitting, which has been shown to have a correlation coefficient of approximately 0.60 with SED fitting using spectroscopic data (Q. Duan et al. 2024). While this moderate correlation does not heavily impact our results, the lack of a large number of spectroscopic observations limits our ability to refine these measurements. If a substantial amount of spectroscopic data were available, we could compute the SFR using Balmer lines and re-analyse the data to verify whether the observed SFR enhancement of 0.26 ± 0.11 at $r_p < 20$ kpc remains consistent, whether the enhancement occurred at larger projection separation values, and whether close-pair samples present a bursty star formation compared to the non-merging sample.

An interesting finding in our study is that the SFR of morphologically identified merging galaxies is lower than that of close-pair galaxies at $r_p < 20$ kpc. This contrasts with previous results. E. A. Shah et al. (2022) found that morphologically identified merging galaxies consistently exhibited higher star formation enhancement than close-pair samples at $0.5 < z < 3.0$. There are three potential explanations for this discrepancy. First, due to the faint and distant nature of high-redshift galaxies, even with *JWST*'s resolution, the morphological parameters used in N. Dalmasso et al. (2024), such as the CAS and Gini coefficients, suffer from high uncertainties, which could lead to falsely identifying merging galaxies that are, in fact, single systems. Moreover, isolated high-redshift galaxies may themselves appear highly disturbed due to internal processes, meaning that the presence of

morphological disturbances alone does not necessarily indicate a merging system. Secondly, it is possible that the majority of morphology-selected galaxies are in the very late stages following coalescence (L. Ferreira et al. 2024). Other works including S. L. Ellison et al. (2022) show that at $0.01 < z < 0.3$, once the merger process is complete, mergers can rapidly quench star formation.

In addition to the observational data, we examine the evolution of star-forming properties using the SC-SAM simulation. In the simulation, we do not detect any significant star formation enhancement at any projected separation from the closest pair. The overall evolution of SFR, sSFR, and stellar mass with projected separation exhibits a largely flat trend, with only minor variations in the smallest bin, $r_p < 20$ kpc, relative to the non-merger median values. However, these deviations are not statistically significant. While the observational stellar mass evolution shows modest alignment with the simulation results, the SFR and sSFR differ considerably. This reflects the fact that close-pair interaction-induced star formation is not explicitly included in the semi-analytic model. Nevertheless, SFR enhancement following mergers (i.e. during post-merger phases) is taken into account, and we find that mergers trigger elevated SFRs, which are subsequently quenched within $\sim 100 - 200$ Myr after coalescence. Conversely, J. M. Scudder et al. (2012) utilized simulations of major mergers developed by P. Torrey et al. (2012b), finding that SFR enhancements at $r_p < 20$ kpc can be as high as 0.6 dex compared to non-merging systems. This demonstrates that in large simulations such as SC-SAM, the detailed processes of galaxy interactions are often underrepresented – possibly because simulating all galaxy-galaxy interactions at very high resolution is computationally expensive. However, if we focus specifically on small-scale galaxy interactions, we are better able to recover and trace the evolution of properties such as SFR as a function of projected separation to the nearest companion.

Lastly, we investigate AGN Fraction and AGN Excess for Type-I and Type-II AGNs at $z = [5.3, 6.7]$ for the first time. We detect an AGN Excess of $1.26^{+0.15}_{-0.04}$ and $1.34^{+0.23}_{-0.11}$ in close-pair samples relative to non-merger samples for AGNs selected using the BPT

diagnostic diagram and SEDs, respectively. Although the excess is not highly significant due to the limited size of our dataset, the consistent detection of AGN Excess from two independent methods strengthens the robustness of these results. One could argue that our Type-I AGNs selected photometric SED methods are less persuasive than direct observations of broad-line (e.g. BL H α AGNs) (e.g. R. Maiolino et al. 2024; I. Juodžbalis et al. 2025; R. Maiolino et al. 2025). However, detecting BL AGNs is another significant task and beyond the scope of this paper. At the time of writing the paper, no comprehensive BL AGN catalogues have been published at these redshifts. Nevertheless, the AGN Excess detected from both the photometric SED and BPT methods lends solid support to the idea that at high-redshift AGNs are triggered in part by the merging process. Please note that for our AGN analysis, we include both major ($\mu > 1/4$) and minor mergers ($\mu < 1/4$). If we were to restrict the analysis to only major mergers, the observed fraction of AGNs with close companions relative to the total AGN population would be lower. Furthermore, differences in the values of the same AGN fraction definition between AGNs identified through SED fitting (Type-I) and those identified via the BPT diagram (Type-II) are expected. These two represent two distinct AGN populations: Type-I AGNs are generally viewed face-on, allowing direct observation of broad emission lines and strong continuum emission; whereas Type-II AGNs are typically viewed edge-on, with the central engine obscured by a dusty torus, resulting in only narrow emission lines being observed. These two distinct AGN populations are not expected to have the same AGN fractions. However, the observed AGN excess in close-pair samples for both AGN populations provides strong evidence that galaxy interactions play a crucial role in triggering AGN activity.

Our analysis highlights how mergers contribute to the evolutionary pathways of galaxies in the early Universe. The detection of SFR enhancement in the closest pairs ($r_p < 20$ kpc) at $z > 4.5$ suggests that mergers are one of the key cornerstones of the elevated star-formation efficiency (SFE) observed at high redshift, alongside other processes such as rapid gas accretion and shorter gas dynamical cooling timescales. In addition, the scatter of the star-forming main sequence (SFMS) has been observed to increase with both redshift and stellar mass (e.g. J. W. Cole et al. 2025; C. Simmonds et al. 2025), with mergers likely playing an important role in driving SFR variability during both the close-pair and post-merger phases. Furthermore, the direct observational evidence of AGN excess in close-pair samples may help explain the elevated AGN population in the early Universe, as well as the extreme luminosity of sources such as GN-z11, which may be fueled by interactions with close companions (S. Tacchella et al. 2023b).

Overall, our results on merger-triggered star formation enhancement and AGN excess are broadly consistent with findings from lower-redshift galaxy studies, showing no significant deviations in the overall trends (e.g. S. L. Ellison et al. 2008; J. M. Scudder et al. 2012; S. Satyapal et al. 2014; E. A. Shah et al. 2022; J. M. Comerford et al. 2024). We find an excess of star formation, that is not quite significant except for the closest galaxies ($r_p < 20$ kpc) at the smallest differences in redshift, between pairs. We also find a significant excess of AGNs in pairs compared to those isolated ones. This excess is only at most a few 10s%, thus less than a factor of two. It remains to be determined how quantitatively this excess results in the growth in super massive black holes. In the future when better data and spectroscopic redshifts are available at these higher redshifts we can determine with more

certainty the role of mergers and galaxies in pairs in producing star formation and AGN activity.

6 CONCLUSIONS

In this paper, we present an early analysis of the impact of mergers on galaxy star-forming properties and AGN activity at $z = 4.5$ – 8.5 . We explore in detail how galaxy SFR, sSFR, and stellar mass evolves as a function of projected separation to the closest neighbour, and whether these properties show any enhancement compared to non-merger samples in both observational and simulation data. Additionally, we examine how the AGN fraction varies between close-pair and non-merger samples and assess the AGN excess in these fractions. For the SFR-enhancement analysis, we restrict our merger sample to pairs with mass ratios greater than 1:4, while no mass ratio constraint is applied for the AGN fraction and AGN excess measurements. Our key findings are summarized as follows:

I. We observe an enhancement in SFR from the observational data at $r_p < 20$ kpc in major merger. The extent of this enhancement is 0.25 ± 0.10 dex above the non-merger median for $z = [4.5, 6.5]$ and 0.26 ± 0.11 dex for $z = [6.5, 8.5]$ (Fig. 3). No enhancement signals are detected at larger separations. We thus conclude that there is a marginal excess in star formation for very close pairs of galaxies.

II. Comparing this work's results with simulations we show that star formation properties derived from the SC-SAM simulation show no significant evolution or variation at $r_p < 100$ kpc. That is, no SFR enhancement signal is detected at $r_p < 20$ kpc in either redshift bin, $z = [4.5, 6.5]$ or $z = [6.5, 8.5]$ (Fig. 4a).

III. The sSFR values in the simulations shows minimal variation at all projected separations and redshift ranges. Stellar mass from the simulations shows a small enhancement of approximately 0.1 dex above the non-merger median at $r_p < 20$ kpc (Figs 4b and c).

IV. We compute the fraction of AGNs that reside in galaxy pairs relative to the total AGN population ($f_{\text{galaxy-pair,AGN}}$) and find values of $0.52^{+0.17}_{-0.24}$ and $0.45^{+0.20}_{-0.14}$ for BPT-selected (Type II) and SED-selected (Type I) AGNs, respectively. This fraction is $3.25^{+1.50}_{-1.06}$ times higher than the corresponding galaxy-pair fraction at the same redshift. Additionally, almost all ($73^{+27}_{-32}\%$) AGNs have at least one companion within 100 kpc (Fig. 8a).

V. We investigate AGN excess in close-pair samples relative to non-merger samples and find an excess of $1.26^{+0.15}_{-0.04}$ for BPT-selected AGNs and $1.34^{+0.23}_{-0.11}$ for photometric SED-selected AGNs. While the excess is not highly significant due to the limited size of our dataset, the consistent detection of AGN excess from two independent methods supports the robustness of these results (Figs 8b and a).

Overall, using currently one of the most extensive dataset from eight *JWST* observational fields (CEERS, JADES GOODS-S, NEP-TDF, NGDEEP, GLASS, El-Gordo, SMACS-0723, and MACS-0416), we explore merger-interaction-driven galaxy star forming and AGN activities at $z = [4.5, 8.5]$ for the first time. This study currently represents one of the highest redshift analysis of mergers and may continue to be one of the most advanced.

A worthwhile future endeavor, when a statistically substantial number of *JWST* spectroscopic observations are available, is to investigate this formation and activity using spectroscopic data e.g. balmer lines for SFR calculations and AGN identification. We are also strongly limited to relatively lower mass galaxies within this study. Ideally we would also like to probe the more massive galax-

ies, but these are very rare at these redshifts. This will require larger surveys with *JWST* or in the future with Euclid/Roman. Likewise, deeper *JWST* surveys are needed to obtain complete samples of galaxies, as at the current time we are incomplete for most masses at the highest redshifts where our distant galaxy sample is measured.

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DATA AVAILABILITY

The data underlying this article is made available by N. J. Adams et al. (2023), D. Austin et al. (2023a), N. J. Adams et al. (2024), D. Austin et al. (2024), T. Harvey et al. (2024), and C. J. Conselice et al. (2024b). The catalogues of the sample discussed herein may be acquired by contacting the corresponding author.

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APPENDIX A: MASS RATIO DISTRIBUTIONS

In this appendix, we present two figures showing the distribution of mass ratios (μ) for our galaxy pairs. The mass ratio is defined as the stellar mass of the less massive galaxy divided by that of the more massive galaxy in the pair. In the first part of the paper, dedicated to studying merger-induced star formation enhancement, we restrict our analysis to major mergers ($\mu > 0.25$). In contrast, the second part, which examines merger-triggered AGN activity, includes all galaxy pairs irrespective of their mass ratio.

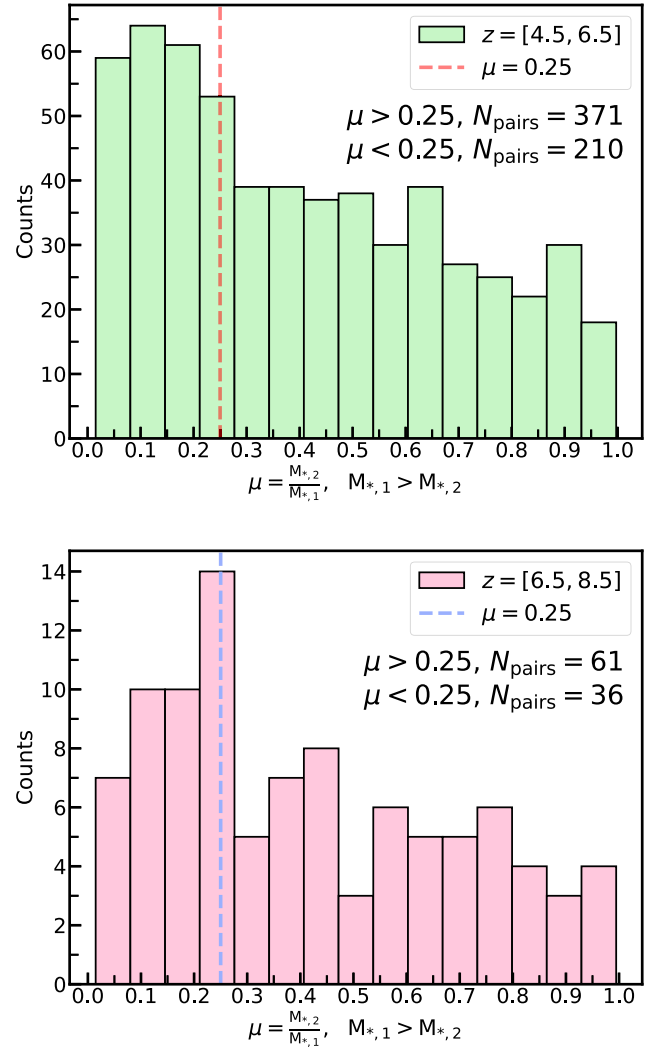


Figure A1. Histograms showing the distribution of merger mass ratios (μ) in our sample across two redshift bins: $4.5 < z < 6.5$ and $6.5 < z < 8.5$, within the stellar mass-complete range $8.0 < \log_{10}(M_*/M_\odot) < 10.0$. The red and blue vertical dashed lines mark $\mu = 0.25$, distinguishing major mergers ($\mu > 0.25$) from minor mergers ($1/10 < \mu < 1/4$). In the first half of the paper, which focuses on merger-induced star formation enhancement, only major mergers are considered. In the second half, which investigates merger-triggered AGN activity, all galaxy pairs are included regardless of their mass ratio.

APPENDIX B: MERGER-TRIGGERED STAR FORMATION ACROSS DIFFERENT LINE-OF-SIGHT CRITERIA

In this appendix, we present the evolution of galaxy SFR, sSFR, and stellar mass as a function of projected separation to the closest neighbour, under different line-of-sight selection criteria. We carry out this test as it is possible that an enhanced signal of star formation would be present at some combination of pair separation and velocity/redshift different. A fiducial $\Delta z < 0.3$ for observational data and $\Delta v < 1000 \text{ km s}^{-1}$ for simulation data were chosen, and our results under these criteria have been presented and discussed in detail in Section 4.1. Here, we explore the variation of these properties using five different Δz upper limits: 0.1, 0.2, 0.3, 0.4, and 0.5 for the observational data, and five different Δv upper limits: 1000, 2000, 3000, 4000, and 5000 km s^{-1} for the simulation data, with results presented in Fig. B1. We also tested results using Δv upper limits of 100, 200, and 300 km s^{-1} for the SC-SAM data, and the results are very similar to those obtained with other Δv upper limits. Overall, we find that the degree of SFR enhancement from observations at $\Delta z < 0.1$ and $r_p < 20 \text{ kpc}$ is more pronounced than the results at other Δz criteria ($\Delta z < 0.2, 0.3, 0.4$). Apart from this, we do not observe significant differences between the results obtained with different line-of-sight criteria.

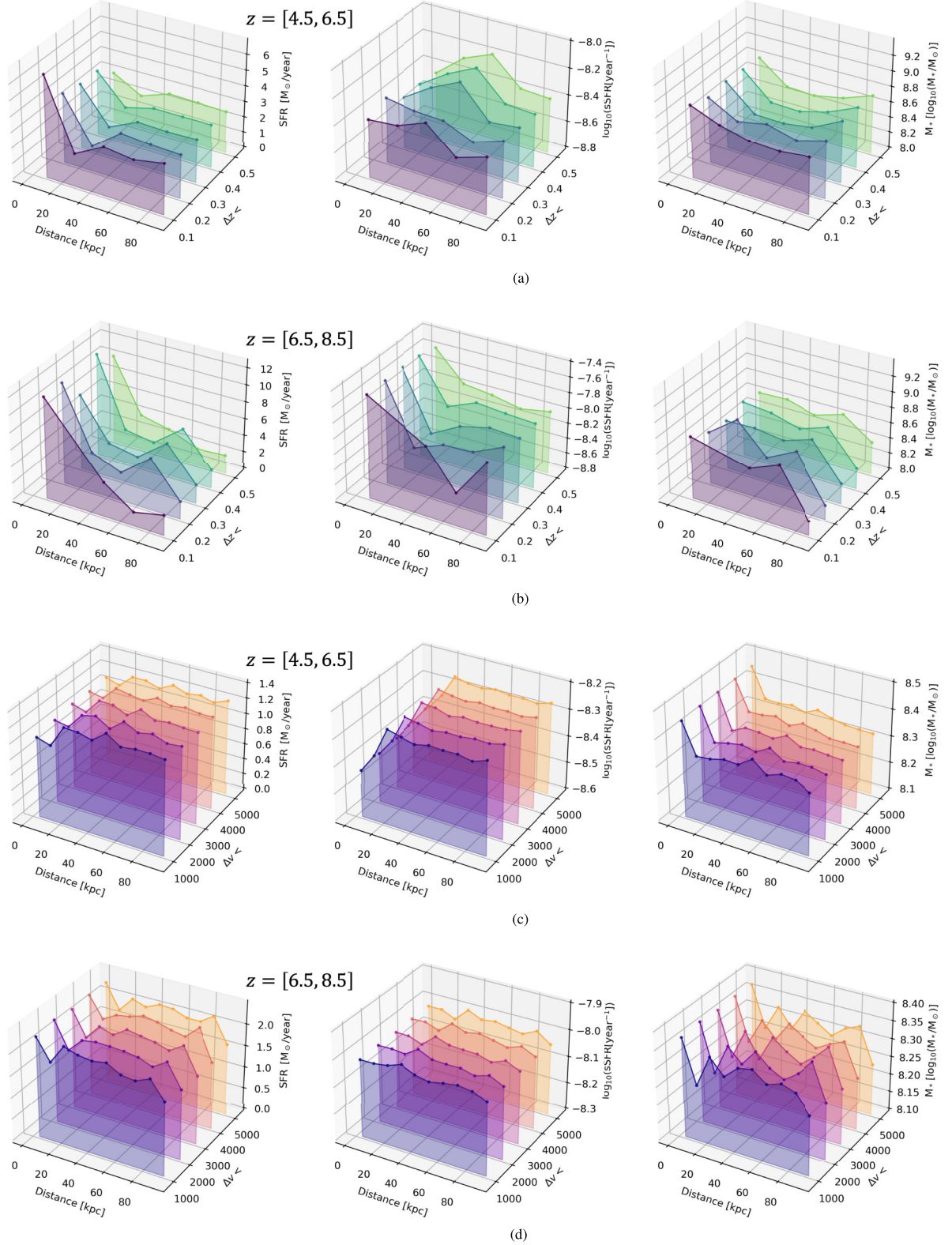


Figure B1. (a) Variation of SFR, sSFR, and stellar mass with projected separation in observational data, using five different Δz upper limit criteria at $z = [4.5, 6.5]$. (b) Variation of SFR, sSFR, and stellar mass with projected separation in observational data, using five different Δz upper limit criteria at $z = [6.5, 8.5]$. (c) Variation of SFR, sSFR, and stellar mass with projected separation in SC-SAM simulation data, using five different Δv upper limit criteria at $z = [4.5, 6.5]$. (d) Variation of SFR, sSFR, and stellar mass as a function of projected separation in the SC-SAM simulation data, using five different Δv upper limit criteria at $z = [6.5, 8.5]$. Four figures illustrating the variation of SFR, sSFR, and stellar mass as a function of projected separation in both observational and simulation data, using different line-of-sight selection criteria.

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